

Mapping when genomic data can and cannot detect malaria transmission: biological constraints on inference

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Abstract

Background. Genomic surveillance is increasingly deployed to track malaria transmission, yet the conditions under which genomic data can reliably detect transmission links remain poorly defined. For *Plasmodium falciparum*, the combined effects of high selfing rates, variable recombination, and elevated background genomic relatedness may fundamentally constrain inference—but this constraint has not been systematically quantified across biologically relevant parameter spaces.

Methods. We developed a forward-time simulation framework using SLiM, tskit, and msprime to generate full-genome sequence data under three scenario classes: a baseline biological parameterisation, a recombination sweep spanning four orders of magnitude ($r = 10^{-9}$ to 10^{-6} per base per generation), and a multi-population structure with migration rates of 0.001–0.05. Transmission detectability was evaluated using three inference approaches—IBD (hmmIBD), identity-by-state (IBS), and maximum-likelihood phylogenetic patristic distances (IQ-TREE)—across varying generation thresholds ($G = 1$ –25), sample sizes ($n = 100$ –400), and population sizes. Performance was quantified using area under the precision-recall curve (AUPRC), F1 score, and sensitivity at fixed specificity. Results were validated against an observed *P. falciparum* dataset using same-household and different-household pairs as a transmission proximity proxy.

Results. Transmission detectability was strongly and primarily determined by recombination rate. AUPRC increased from 0.53–0.62 at $r = 10^{-9}$ to 0.74–0.82 at $r = 10^{-6}$ regardless of migration rate or sample size. IBD-based inference consistently outperformed IBS and phylogenetic approaches, though the absolute advantage was small. The generation threshold G was a critical secondary modulator: at $G = 1$, no method exceeded AUPRC = 0.50 under any recombination condition. Migration rate exerted minimal independent influence on detectability across the range explored. In observed *P. falciparum* data, all three methods performed near the random classifier baseline (AUPRC < 0.05), consistent with near-complete IBD distribution overlap between household-linked and unlinked pairs.

Conclusions. Transmission detection from genomic data in *P. falciparum* is subject to fundamental biological constraints that precede methodological choices. Recombination rate and the temporal depth of the transmission definition are the primary determinants of whether genomic inference can

succeed. Current IBD and IBS tools applied to endemic African populations operate in a parameter regime where accurate transmission detection is unlikely without integration of epidemiological and spatial data. These results provide a quantitative framework for calibrating expectations for genomic surveillance programmes and identifying conditions under which genomic data alone is insufficient to resolve transmission connectivity.

Keywords: *Plasmodium falciparum*; malaria genomic epidemiology; IBD; transmission inference; simulation; recombination; detectability; hmmIBD; IQ-TREE; AUPRC

Introduction

Understanding how malaria spreads between individuals and across communities remains central to malaria control and elimination (World Health Organization 2023). Transmission dynamics determine where interventions should be targeted, how rapidly parasites move through populations, and whether local transmission persists despite control efforts (Smith et al. 2010; Griffin et al. 2010). Over the past decade, genomic data have become an increasingly integral component of malaria surveillance, applied to distinguish local transmission from importation, map parasite population connectivity, and reconstruct transmission networks at fine spatial and temporal scales (Neafsey, Juraska, et al. 2015; Wesolowski et al. 2018; MalariaGEN et al. 2021). Large-scale sequencing initiatives have demonstrated that genome-wide variation captures relatedness gradients reflective of recent shared ancestry, enabling probabilistic inference of epidemiological connectivity across diverse endemic settings (Daniels et al. 2015; Alyssa R. Taylor et al. 2017; Zhu et al. 2022; Diez Benavente et al. 2020).

Despite these advances, a critical and underexamined question persists: under what biological and surveillance conditions can the genomic signal of recent transmission actually be detected? Most genomic transmission studies proceed under an implicit assumption that sufficient data collection and appropriate analytical tools will recover meaningful transmission signal (Didelot et al. 2017; Campbell et al. 2018). Yet *P. falciparum* poses a uniquely challenging inferential environment. Obligate sexual recombination during reproduction in the mosquito midgut progressively erodes haplotype-level relatedness between co-transmitted parasite lineages, fragmenting shared ancestry into increasingly short and sparse genomic segments with each successive transmission cycle (Conway et al. 1999; Jiang et al. 2011; Miles et al. 2016). Polyclonal infections, in which a single human host carries genetically distinct parasite clones, further obscure pairwise genomic signals by introducing background relatedness that is difficult to separate from transmission-derived similarity (Nkhoma et al. 2020; Felger et al. 2012). In endemic settings, high background relatedness arising from population structure and ongoing gene flow compounds these difficulties (Ferreira et al. 2007; Chenet et al. 2012; Amambua-Ngwa et al. 2019). Together, these features suggest that the limits of transmission signal detection in malaria are not merely methodological, but biological.

The critical gap in the existing literature is the absence of a principled characterisation of when the genomic signal of recent transmission transitions from detectable to undetectable, independent of the inference approach applied. Prior work has established that genomic relatedness statistics can capture epidemiological connectivity under favourable conditions (Daniels et al. 2015; Aimee R. Taylor et al. 2017; W. Wong et al. 2017; Henden et al. 2018), and that their performance degrades under high recombination or complex population structure (Schaffner et al. 2018; Speed and Balding 2015). However, these studies evaluate performance relative to inferred proxies or incomplete empirical benchmarks, without access to known transmission ground truth, making it difficult to determine whether inference failure reflects signal absence or analytical inadequacy (Didelot et al. 2021; Albers and McVean 2020). As a result, the parameter space in which transmission detection is biologically feasible and the boundaries where it collapses regardless of analytical choice remains uncharacterised.

Here, we develop a simulation-based framework to map the detectability landscape of recent *P. falciparum* transmission connectivity across the key biological and surveillance dimensions that govern signal availability. Using forward-time simulations with complete genealogical ground truth (Haller and Messer 2019; Kelleher et al. 2018), we define transmission-linked ancestry precisely and quantify how it is transformed, preserved, degraded, or rendered indistinguishable from background as recombination rate, population structure, sampling intensity, and transmission window depth vary. Rather than comparing inference methods as competing approaches, we deploy three established genomic summary statistics, identity by descent estimated via hmmIBD (Schaffner et al. 2018), identity by state (IBS), and phylogenetic inference from maximum-likelihood trees (T. Wong et al. 2025), as complementary instruments for probing the boundaries of detectable signal. Consensus failure across all three instruments is treated as evidence of a biological detectability limit, not a methodological one (Baumdicker et al. 2022; Speed and Balding 2015).

By mapping when genomic data can and cannot recover evidence of recent transmission connectivity, this study reorients malaria genomic epidemiology around a question that precedes the choice of analytical method: does the signal exist? The boundaries delineated here define realistic expectations for genomic transmission surveillance, clarify the conditions under which investment in sequencing and analysis is likely to return informative results, and provide a biologically grounded

framework for interpreting relatedness patterns in field-collected parasite genomes. Rather than proposing a new inference method, we establish the biological parameter space within which any method can succeed and quantify how rapidly that space contracts as recombination, sampling constraints, and transmission complexity increase.

A staged simulation framework for quantifying genomic identifiability limits under controlled biological and surveillance variation

In malaria, transmission occurs through *Anopheles* mosquitoes whose movements and infection histories are not individually tracked. This vector-mediated biology means that genomic ancestry observed in human infections reflects population-level mixing rather than discrete person-to-person transmission chains, rendering direct reconstruction of “who infected whom” from sequence and/or epidemiological data largely intractable. Rather than attempting to recover individual transmission events, the analyses that follow quantify the conditions under which genomic data can accurately recover transmission linked ancestry across varying biological and surveillance constraints. To this end, we constructed a simulation framework coupling three scenario blocks, a baseline biological regime, a recombination rate sweep, and a structured multi-population demography with a staged genomic simulation pipeline and a multi-method inference layer evaluated against ground-truth IBD using precision, recall, F1 and area under the precision-recall curve (AUPRC) (**Figure 1**). This design ensures that observed limits on recoverability are attributable to biological and demographic parameters rather than to differences in analytical procedure. Simulation parameters for all three scenarios are summarized in **Supplementary Table S1**. The bottleneck effect on inference performance is characterized separately in **Supplementary Figure S1**, which shows that variation in effective population size at the demographic bottleneck ($N_e = 1,000$ to 20,000) produces AUPRC differences of 0.011–0.014 across methods at medium recombination, within the confidence interval range and substantially smaller than the effects attributable to recombination rate variation.

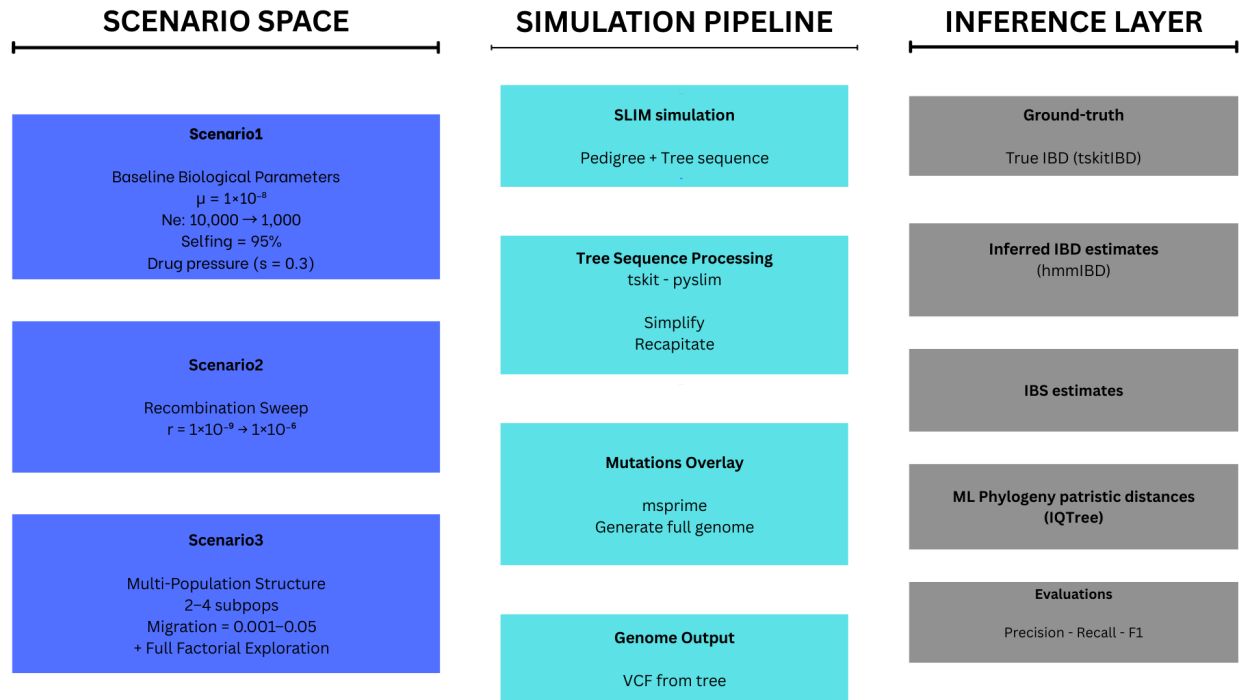


Figure 1. Overview of the simulation scenarios, genomic data generation, and inference evaluation framework used for transmission evaluation. The figure summarizes the end-to-end framework linking simulated epidemiological scenarios to genomic inference and performance evaluation. The figure outlines the conceptual and computational structure used to evaluate limits of genomic transmission identifiability. The left column defines the scenario space explored, including a baseline biological setting, a recombination-rate sweep, and structured multi-population models with migration. The central column depicts the simulation pipeline, from forward-time simulations with pedigree and tree-sequence recording, through tree-sequence processing and mutation overlay, to generation of genome-wide variant data. The right column summarizes the inference layer, distinguishing between ground-truth ancestry (true identity by descent (IBD)), genomic summaries derived from observable data (inferred IBD, identity by state (IBS), and phylogenetic cophenetic distances from maximum-likelihood trees), and downstream evaluation framework based on precision, recall, and F1 metrics.

Results

1. Transmission-linked parasite pairs retain only partial genealogical separation, establishing a biological ceiling on detectable signal.

Before any inference layer is applied, the most fundamental question for transmission detectability is whether the biological system produces meaningfully distinct genealogical signals between directly linked and unrelated parasite pairs. We addressed this by examining the distribution of true coalescence times (TMRCA) between transmission-linked pairs (direct descendants within a defined generational window, $G \leq 25$) and epidemiologically unrelated pairs (non-direct) under baseline simulation parameters.

A prerequisite for any genomic method to distinguish transmission-linked pairs from unlinked pairs is that the two classes occupy separable regions of the genomic signal space. We examined this prerequisite using the distribution of true pairwise coalescence times (TMRCA) under the baseline and recombination sweep scenarios (**Figure 2**). In both scenarios, direct transmission pairs (orange) display a distribution shifted toward shorter TMRCA values relative to non-direct pairs (blue), consistent with more recent common ancestry. However, the distributions overlap considerably across the full range of TMRCA values observed, with non-direct pairs contributing substantial density at low TMRCA values in both scenarios.

Under the baseline scenario, direct pairs cluster below approximately 50 generations, but the tail of the non-direct distribution extends well into this range, producing a region of irreducible overlap. This overlap is at least as pronounced in the recombination sweep scenario, where both distributions appear broader and their modes more similar. The degree of overlap in TMRCA space does not markedly increase with elevated recombination rate alone, suggesting that the biological challenge of separation is present even before accounting for the additional distortion introduced by the inference process. Together, these distributions establish that the genomic signal separating direct from non-direct transmission is inherently limited in this pathogen system, and that any classifier operating on this signal will face a non-trivial false positive burden regardless of method choice.

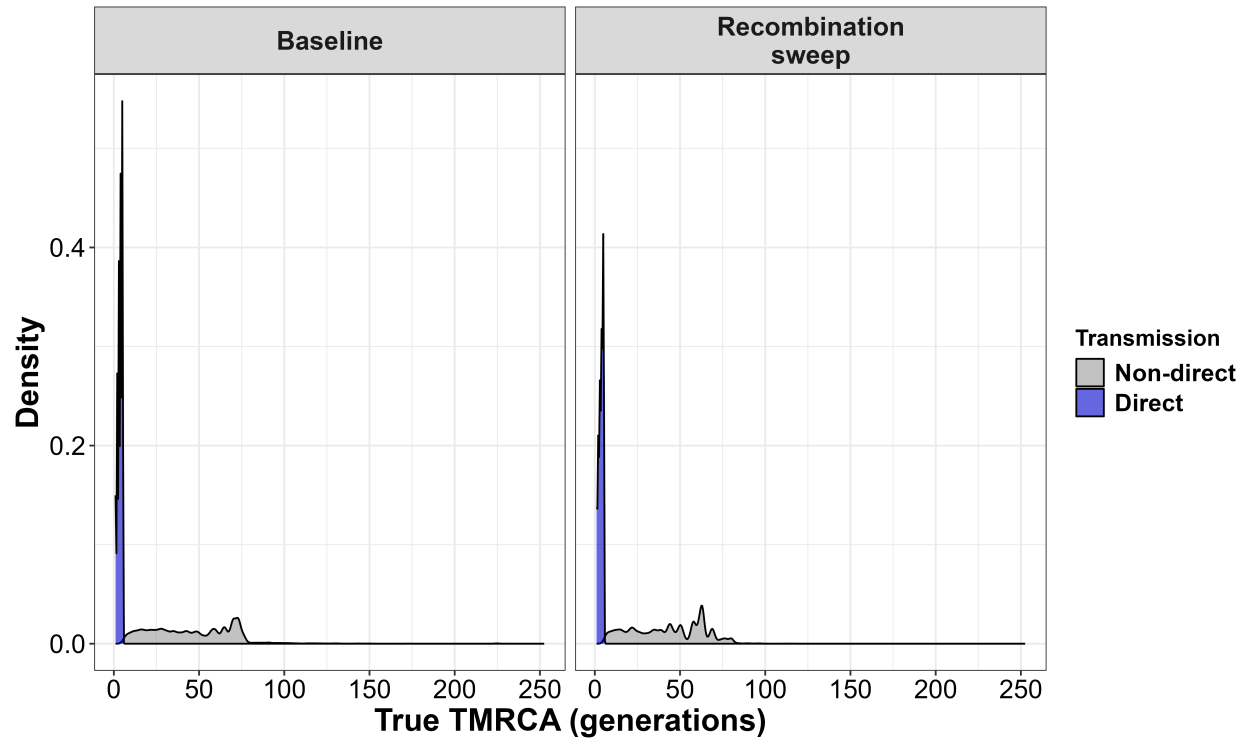


Figure 2: True TMRCA distributions for direct (orange) and non-direct (blue) transmission pairs under the baseline scenario (left panel) and recombination sweep scenario (right panel). The x-axis shows true TMRCA in generations; the y-axis shows density. Each panel pools all replicate simulations under the corresponding scenario. Distributions are shown as kernel density estimates.

2. Recombination progressively erodes the genomic signal of shared ancestry, widening the gap between available and recoverable transmission events.

We quantified how recombination rate modulates both the true genomic relatedness signal and the capacity of hmmlBD to recover it. Across a four-order-of-magnitude range of recombination rates ($r = 10^{-9}$ to 10^{-6} per base per generation), true pairwise IBD proportions declined monotonically with increasing recombination (**Figure 3A**). At $r = 10^{-9}$, the distribution of true IBD is bimodal, with a substantial fraction of pairs exhibiting IBD proportions approaching 1.0, consistent with near-complete identity in very low-recombination genomes (**Figure S1**). As recombination rate increases to $r = 10^{-6}$, this high-IBD mode collapses and the distribution concentrates at lower values, reflecting the progressive fragmentation of identity blocks by recombination.

The inferred IBD distribution from hmmlBD (**Figure 3B**) tracks the true IBD distribution qualitatively across recombination rates, but with systematic compression at intermediate values. The mean inferred IBD proportion parallels the decay of true IBD, though consistently underestimates it at low recombination rates and shows convergence at higher rates (**Figure 3C**). The correlation between true and inferred IBD decreases substantially with increasing recombination (Figure 3D), declining from approximately 0.9 at $r = 10^{-9}$ to roughly 0.55–0.60 at $r = 10^{-6}$. This progressive deterioration in correlation indicates that the informational fidelity of hmmlBD is itself a function of the recombination landscape, becoming increasingly unreliable as recombination rate rises. The implication is that detectability of transmission is partly constrained upstream of any classification step, by the quality of the IBD signal that can be extracted from the genome.

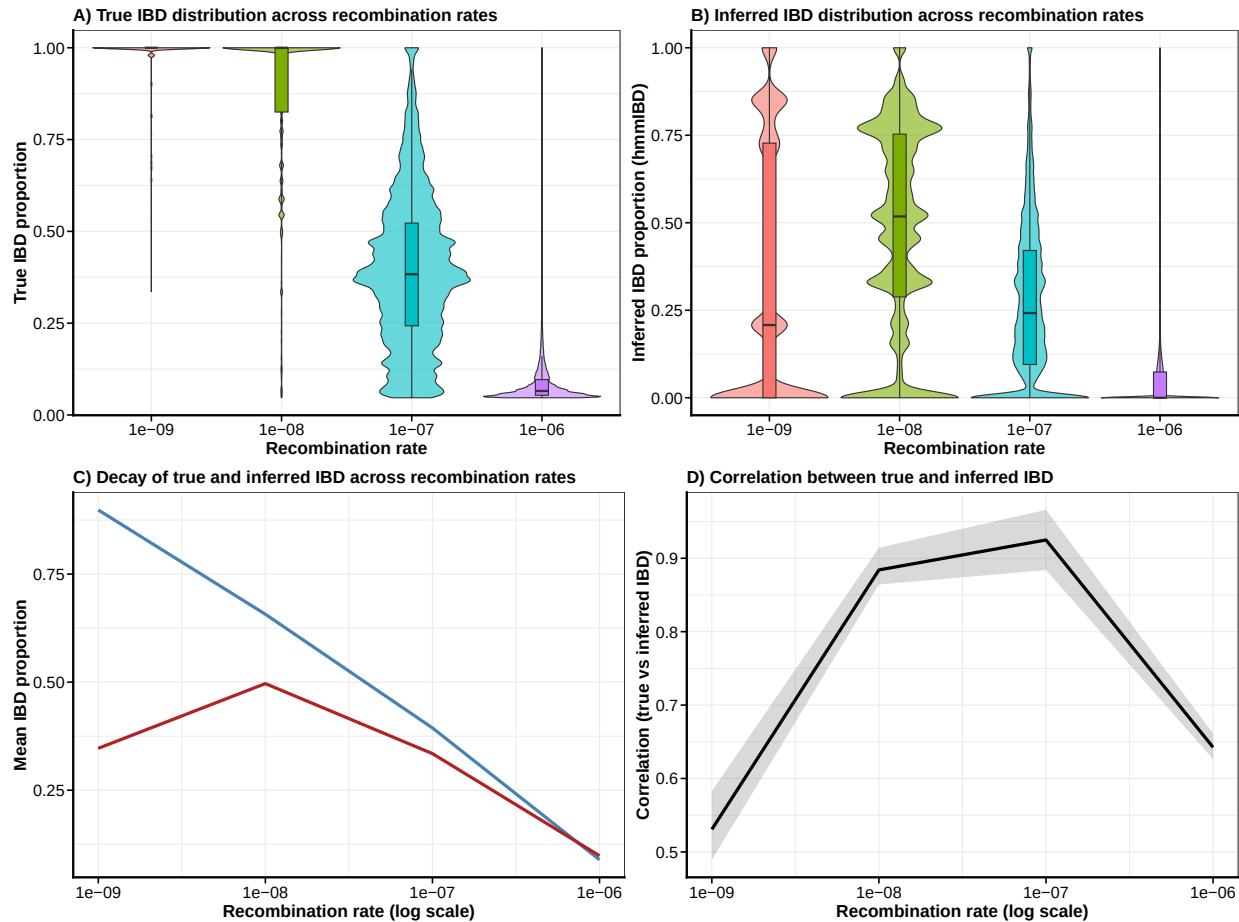


Figure 3: IBD signal decay and inference fidelity across recombination rates. (A) Distribution of true pairwise IBD proportions for recombination rates spanning $r = 10^{-9}$ to $r = 10^{-6}$, shown as

violin plots. (B) Distribution of IBD proportions inferred by hmmlBD across the same recombination rates. (C) Mean true (solid) and mean inferred (dashed) IBD proportions as a function of recombination rate on a log scale, with 95% confidence intervals. (D) Pearson correlation between true and hmmlBD-inferred pairwise IBD proportions as a function of recombination rate (log scale).

3. Transmission detection undergoes a threshold transition governed by recombination rate, independent of inference approach.

Precision-recall profiles and F1 scores varied substantially across inference methods (IBD, IBS, phylogenetic patristic distances), scenarios (baseline, recombination sweep, migration), and recombination rates (**Figure 4**). In the precision-recall space (**Figure 4A**), all three methods show a wide dispersion of operating points, with no method achieving uniformly high precision and recall simultaneously across scenarios. IBD tends to cluster at higher precision values, while IBS and phylogenetic-based inference show broader scatter toward lower recall. The migration scenario introduces the most heterogeneity, with operating points spread across a larger region of the precision-recall plane compared to the baseline and recombination sweep.

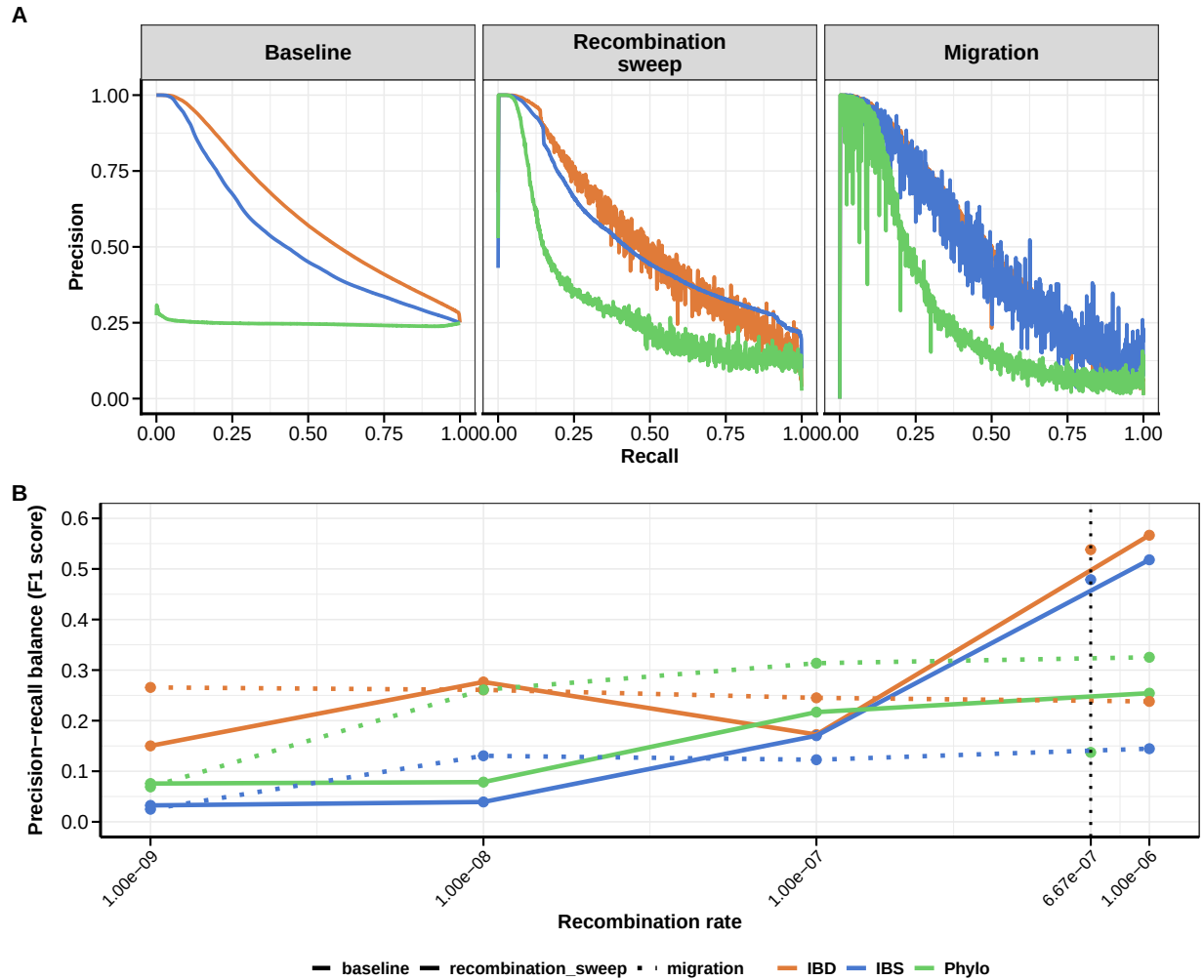


Figure 4. Classification performance of IBD, IBS, and phylogenetic (Phylo) inference methods across simulation scenarios. (A) Precision-recall scatter plot for all scenario-method combinations. Each point represents one replicate simulation. Colour indicates scenario (baseline, recombination sweep, migration); symbol shape indicates method. (B) F1 score as a function of recombination rate for each method, stratified by scenario. Lines show mean F1; shaded regions show 95% confidence intervals across replicates. The x-axis is on a log scale.

F1 scores across recombination rates (**Figure 4B**) reveal a non-monotonic relationship in several scenario-method combinations. Under the recombination sweep, F1 scores for all methods tend to be lower at the extremes of the recombination range and modestly higher at intermediate values, though the differences are small in absolute terms. The baseline and migration scenarios display distinct F1 trajectories, with the migration scenario generally yielding lower scores. Across all

recombination rates, no method consistently exceeds an F1 of approximately 0.6, and many conditions yield F1 scores below 0.4. The method comparison across sensitivity and false discovery rate (Figure Sx) confirms that IBD achieves the most favourable sensitivity-FDR trade-off across scenarios, whereas IBS and phylogenetic inference accumulate higher false discovery rates at comparable sensitivity levels, particularly under the migration scenario.

The cophenetic (patristic) distance distributions for direct and non-direct pairs (Figure 5) provide additional context for the phylogenetic method's performance. In both the baseline and recombination sweep scenarios, the distributions of cophenetic distances for direct and non-direct pairs overlap broadly. The degree of separation between the two classes is modest in the baseline scenario and does not improve in the recombination sweep, where elevated recombination reshuffles ancestry and blurs phylogenetic signal. The limited discriminative capacity visible in the distance distributions is broadly consistent with the phylogenetic method's relatively low F1 and precision-recall performance observed in Figure 4.

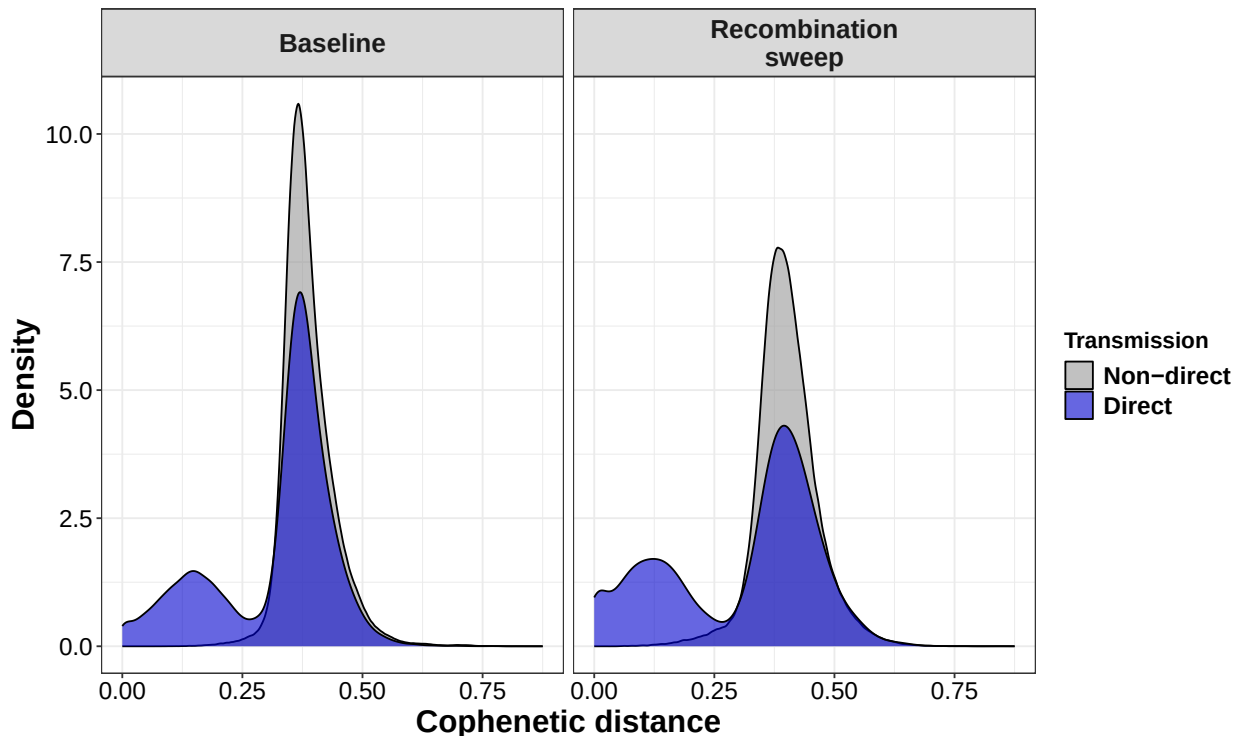


Figure 5: Cophenetic (patristic) distance distributions for direct (orange) and non-direct (blue) transmission pairs under the baseline (left panel) and recombination sweep (right panel) scenarios.

Distances were derived from maximum-likelihood phylogenies inferred with IQ-TREE. The x-axis shows cophenetic distance; the y-axis shows density. Distributions are shown as kernel density estimates pooled across replicates.

4. Migration rate has a limited independent effect on detectability relative to recombination.

To disentangle the contributions of population structure and recombination to transmission detectability, we evaluated AUPRC across a factorial combination of recombination rates (10^{-9} to 10^{-6}) and migration rates (0.001, 0.01, 0.05) for $G \leq 25$ generations (**Figure 6**). Across all panels (**Figure 6A–D**), AUPRC increased with recombination rate for all three methods, regardless of migration level. IBD consistently achieved the highest AUPRC among the three methods at every recombination-migration combination tested. The gradient across migration rates within any fixed recombination rate was modest: at $r = 10^{-6}$ (**Figure 6D**), AUPRC for IBD varied by less than 0.05 units across the three migration levels. The AUPRC gain curve (**Figure 6E**) confirms that the primary driver of detectability differences across the panel is recombination rate, not migration.

At fixed recombination rates, the sensitivity at 90% specificity (**Figure S3**) showed a similar pattern: IBD achieved the highest sensitivity across all migration-recombination combinations, but migration rate did not produce consistent directional changes in sensitivity within any recombination stratum. The AUPRC profiles stratified by migration rate (**Figure S6**) further demonstrate that, at the AUPRC = 0.6 reference threshold, only the highest recombination rate ($r = 10^{-6}$) reliably produces AUPRC values above this level for IBD, and only under some migration conditions. IBS and Phylo remained below this threshold across nearly all conditions. Sensitivity at 90% specificity as a function of migration rate (**Figure Sxxx**) shows that increasing migration from 0.001 to 0.05 does not substantially alter sensitivity in a consistent direction for any method or recombination rate, suggesting that the range of migration rates explored here represents a regime where connectivity does not strongly modulate pair-level discriminability.

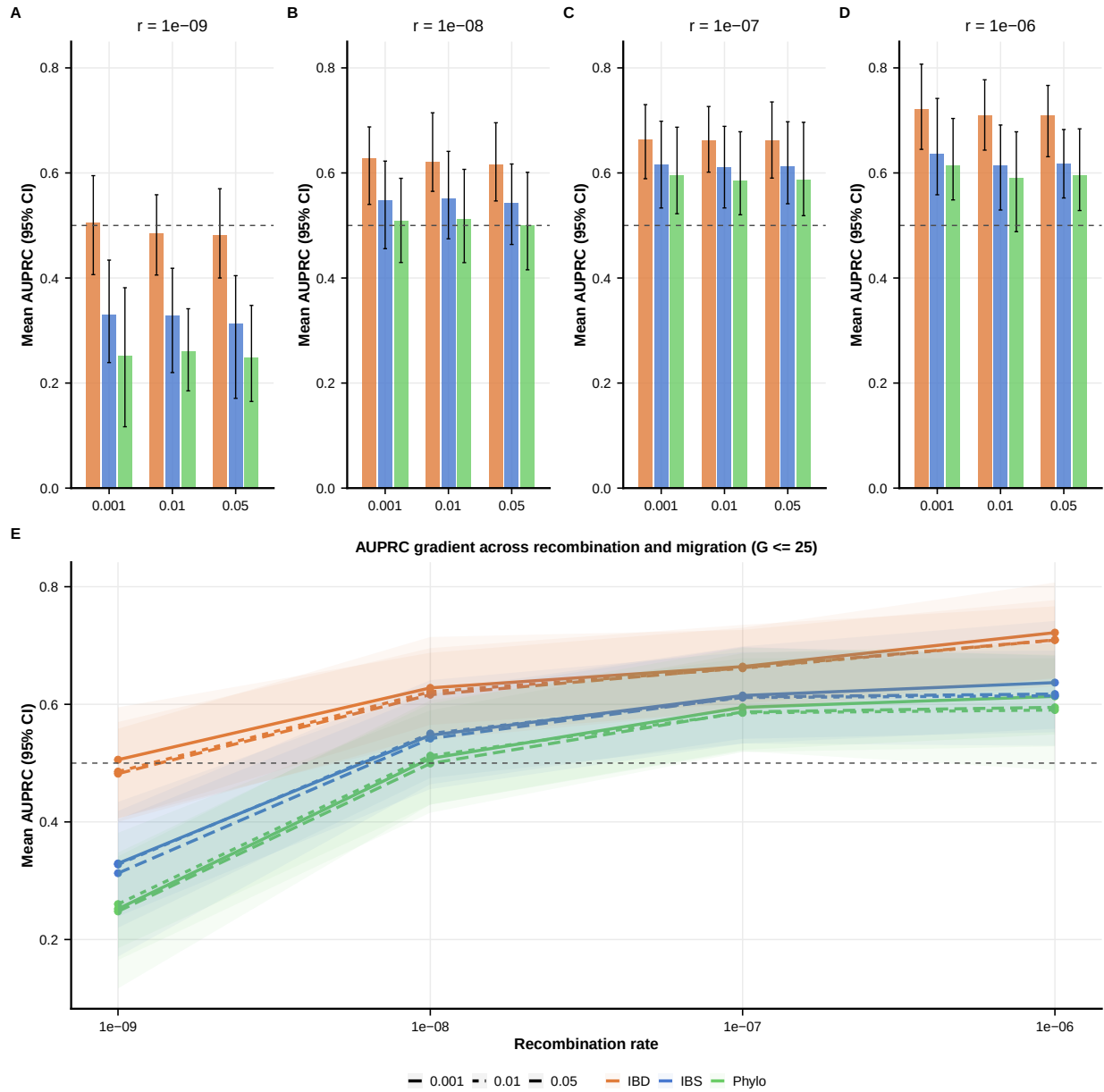


Figure 6: Mean AUPRC ($\pm$ 95% CI) for IBD (orange), IBS (blue), and Phylo (green) methods across migration rates (0.001, 0.01, 0.05) and recombination rates ($r = 10^{-9}$ to 10^{-6}), for pairs within $G \leq 25$ generations. Panels A–D show AUPRC for each recombination rate separately, with migration rate on the x-axis. Panel E shows the AUPRC gradient across all recombination rates simultaneously, with the three migration levels represented by line colour. A horizontal reference line at AUPRC = 0.6 is included.

5. Surveillance design shapes detectability within the biological boundary but cannot transcend it.

To characterise the joint effect of recombination rate, sample size, and migration on the upper bound of detectability, we summarised the best AUPRC achievable by any single method (maximum over IBD, IBS, and Phylo) across a grid of simulation conditions (**Figure 7**). The heatmaps reveal a consistent gradient along the recombination axis: best AUPRC increased from 0.53–0.62 at $r = 10^{-9}$ to 0.74–0.82 at $r = 10^{-6}$, regardless of migration rate or sample size. This gradient was present across all three migration panels (migration = 0.001, 0.01, 0.05) and was the dominant source of variation in the heatmaps.

The effect of sample size on best AUPRC was non-monotonic and generally small relative to the recombination effect. At several recombination-migration combinations, the smallest sample size ($n = 100$) achieved the highest best AUPRC within its column for example, $n = 100$ yielded best AUPRC of 0.82 at $r = 10^{-6}$, migration = 0.001, compared to 0.75 at $n = 400$ under the same conditions. This pattern was not universal and was less consistent at lower recombination rates. Across migration levels, the best AUPRC values were broadly similar: the difference between migration = 0.001 and migration = 0.05 at any fixed recombination-sample size cell was generally less than 0.05 AUPRC units. These results reinforce the conclusion that recombination rate is the primary axis of variation in transmission detectability within the parameter space explored here, while sample size and migration exert secondary and context-dependent effects.

Across the entire grid, best AUPRC did not reach 0.9 in any cell, and only a small subset of conditions, specifically those combining the highest recombination rate with smaller sample sizes, exceeded 0.8. This ceiling on best achievable performance, even under the most favourable simulated conditions, indicates a fundamental limit to transmission detectability that persists despite method optimisation.

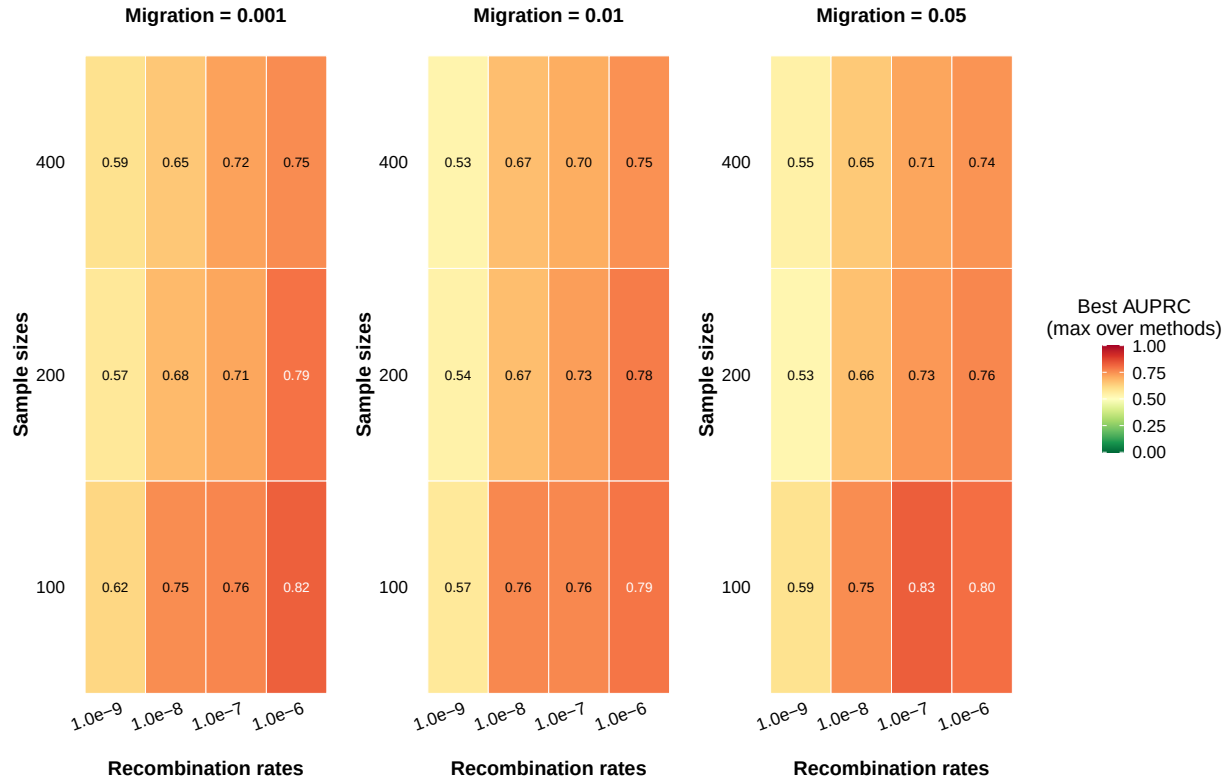


Figure 7: Best AUPRC (maximum over IBD, IBS, and Phylo methods) across sample sizes (y-axis: 100, 200, 400) and recombination rates (x-axis: 10^{-9} , 10^{-8} , 10^{-7} , 10^{-6}), for three migration rates (panels: 0.001, 0.01, 0.05). Colour scale ranges from yellow (low AUPRC ≈ 0.50) to dark red (high AUPRC ≈ 1.00). Numeric AUPRC values are displayed in each cell. Results are for $G \leq 25$ generations.

6. The generation threshold for transmission definition is a critical and underappreciated modulator of detectability.

The definition of what constitutes a “transmission link” depends on the temporal scope specified by the generation threshold G . We systematically varied G from 1 to 25 generations and evaluated AUPRC for each method across all recombination rates (**Figure 8**). At $G = 1$, the most stringent definition of transmission, AUPRC values were low and nearly uniform across recombination rates for all three methods, ranging from approximately 0.39 to 0.46 (**Figure 8A–C**). At this threshold, no method achieved AUPRC above 0.50 under any recombination condition, indicating that genomic

inference of the most recent transmission events is near-random in terms of ranking performance.

As G increased, AUPRC improved for all methods, but the rate of gain was strongly modulated by recombination rate. At $r = 10^{-6}$, AUPRC for IBD increased from 0.41 at $G = 1$ to 0.71 at $G = 25$, a gain of 0.30 AUPRC units (**Figure 8A**). The corresponding gain for IBS was from 0.39 to 0.62 (**Figure 8B**), and for Phylo from 0.42 to 0.60 (**Figure 8C**). By contrast, at $r = 10^{-9}$, AUPRC increases with G were smaller and less consistent across methods, IBD increased from 0.42 to 0.49, while IBS and Phylo showed negligible gains. The G -threshold gain curve (**Figure 8D**) illustrates this recombination- G interaction: the curve is steepest at the highest recombination rates, while at the lowest recombination rate, the AUPRC gain with G is nearly flat.

The G -threshold heatmaps for all three methods (**Figure S7**) confirm these patterns at the replicate level, showing that the IBD method consistently achieved the highest AUPRC at each G -recombination combination, with the largest absolute advantage over IBS and Phylo at $G = 25$ and high recombination. These results indicate that the detectability of transmission events is fundamentally conditional on how broadly “transmission” is defined temporally. Broader definitions (larger G) yield more detectable signals, particularly in genomes with higher recombination, whereas narrow definitions, most relevant for outbreak or household-level epidemiology, remain poorly recovered by current methods across the full recombination range explored.

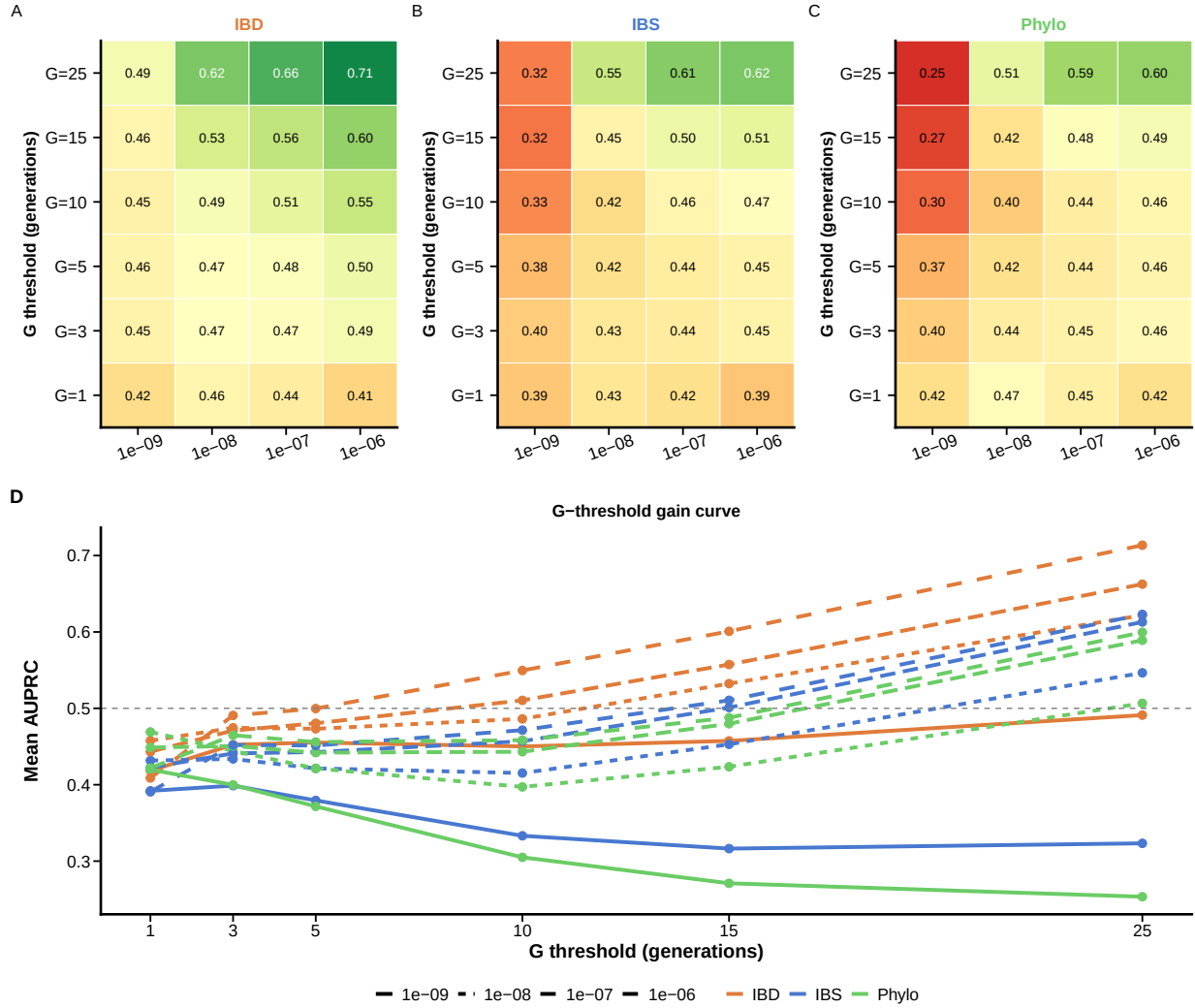


Fig 8: AUPRC as a function of generation threshold G (y-axis: $G = 1, 3, 5, 10, 15, 25$) and recombination rate (x-axis: $10^{-9}, 10^{-8}, 10^{-7}, 10^{-6}$), for IBD (panel A), IBS (panel B), and Phylo (panel C). Numeric AUPRC values are shown in each cell; colour scale from light yellow (low) to dark orange/red (high). Panel D shows the G-threshold gain curve: mean AUPRC as a function of G threshold, with one line per recombination rate and line type per method (IBD, IBS, Phylo).

7. The detectability landscape defines a constrained but navigable parameter space for malaria genomic surveillance.

To contextualise simulated performance against real sequencing data, we evaluated IBD, IBS, and phylogenetic inference using an observed *P. falciparum* dataset in which same household

(same-HH) and different household (different-HH) pairs served as a proxy for transmission proximity. Observed AUPRC values for all three methods (**Figure 9B**) were low, approximately 0.03–0.05, placing them far below the simulated performance distributions (mean simulated AUPRC: ~0.55–0.65 for IBD, ~0.40–0.50 for IBS, ~0.25–0.35 for Phylo). The observed data points fall near the dashed vertical line marking a random classifier (AUPRC $\approx$ prevalence of positive pairs), indicating that none of the three methods could meaningfully distinguish same-HH from different-HH pairs in this dataset.

The IBD proportion distributions for same-HH ($n = 158$) and different-HH ($n = 12,562$) pairs (**Figure 9C**) provide a direct explanation for this failure: both distributions are nearly superimposable, with peaks near IBD proportion ≈ 0.22 – 0.25 . Same-HH pairs show a marginally heavier tail at higher IBD proportions, but the overlap is extensive and the distributions cannot be separated at any reasonable threshold. The dashed vertical reference line in panel C indicates that the bulk of both populations clusters within a narrow range of IBD values, reflecting a population where background relatedness is high and within-household transmission pairs are not detectably more related than random pairs at the scale of pairwise IBD.

Overlaying the observed data AUPRC on the simulated heatmap (**Figure 9A**) shows that the observed performance falls within the region of the parameter space corresponding to low recombination rates and the associated low-detectability zone. The dashed contour lines in the migration = 0.001 and migration = 0.05 panels indicate that the observed data are broadly consistent with simulations in the $r = 10^{-7}$ to $r = 10^{-6}$ range at moderate to large sample sizes and show AUPRC greater than the chosen threshold of 0.8, though the mismatch between observed and simulated AUPRC suggests that additional factors such as high background relatedness, sample aggregation structure, or the imprecision of household membership as a transmission proxy, further reduce performance beyond what the simulation captures.

Sensitivity analysis of the bottleneck size (**Figure S2**) showed that AUPRC varied modestly with the effective population size at the bottleneck (N_e bottleneck = 1,000–20,000) for both low ($r = 10^{-9}$) and medium ($r = 10^{-8}$) recombination rates. The smallest bottleneck produced marginally higher AUPRC, particularly for IBD at low recombination, consistent with a more recent coalescent and stronger relatedness signal under severe demographic contraction. However, the absolute dif-

ferences across bottleneck sizes were small, indicating that the parameterisation of the bottleneck does not qualitatively alter the conclusions about detectability limits.

The fraction of simulation replicates in which transmission events were classifiable as identifiable—defined as exceeding a high-performance AUPRC threshold—was near zero for the large majority of conditions (Figure S4). Only IBD at $r = 10^{-9}$ achieved non-trivial identifiable fractions (0.70–0.95 depending on migration level), and this fraction dropped sharply to below 0.15 at $r = 10^{-8}$. IBS and Phylo methods yielded identifiable fractions of 0.00 across all recombination rates and migration levels, underscoring the near-complete failure of these methods to achieve high-confidence transmission classification in this pathogen system. Finally, the AUPRC gap between IBD and the next-best method (**Figure S5**) was largest at the lowest recombination rate (gap = 0.158–0.176 across migration levels) and smallest at $r = 10^{-7}$ (gap $\approx$ 0.049–0.050), indicating that the three methods are most differentiated in the regime where the genomic signal is strongest. Across all conditions, IBD retained a positive advantage over IBS and Phylo, though this advantage provides limited practical benefit when absolute AUPRC values remain below 0.6 in most conditions.

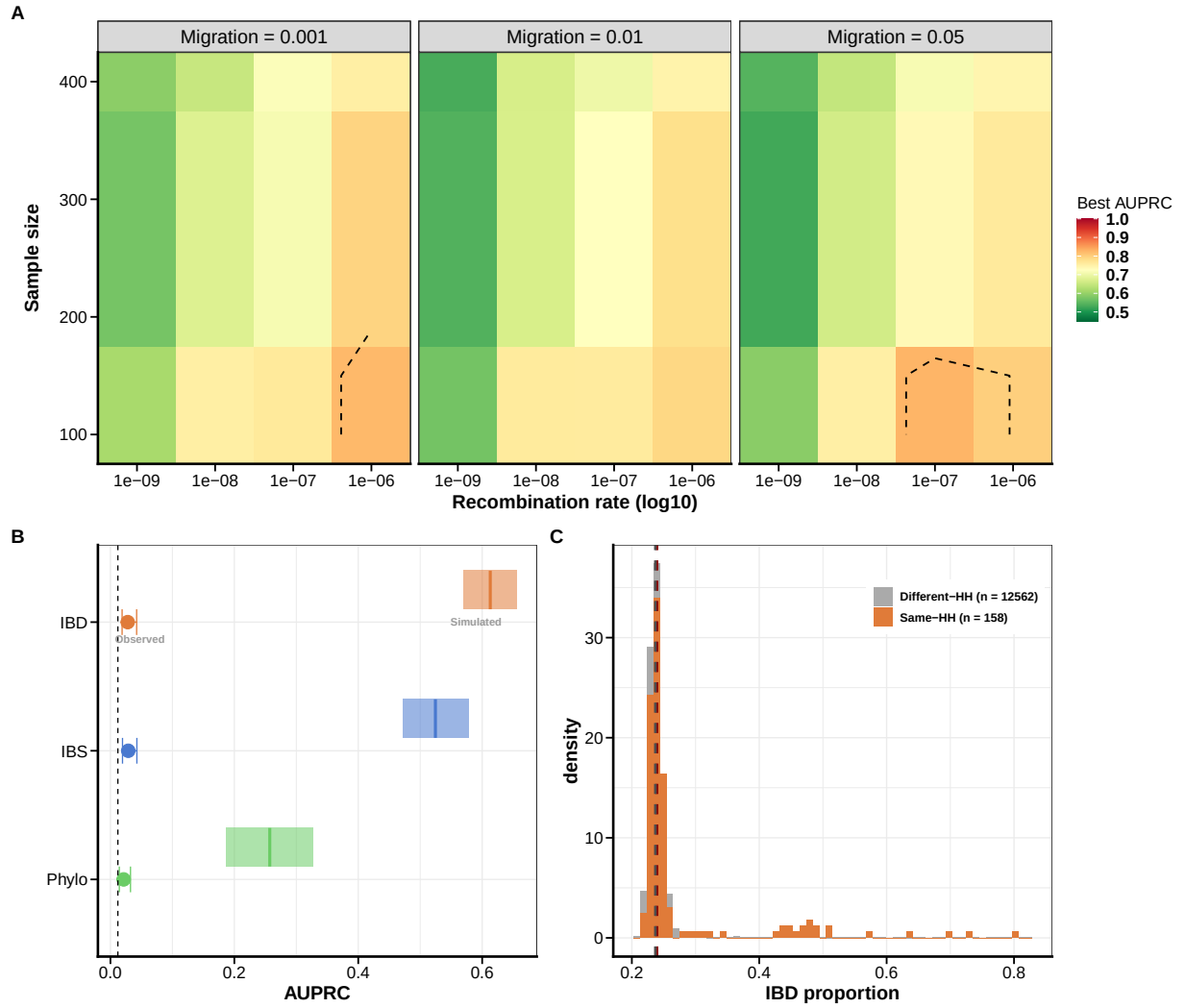


Figure 9: (A) Heatmap of best simulated AUPRC (maximum over IBD, IBS, Phylo) as a function of recombination rate (x-axis, log scale) and sample size (y-axis: 100–400), for three migration rates (panels: 0.001, 0.01, 0.05). Dashed contour lines indicate the region of the parameter space most consistent with observed *P. falciparum* data. Colour scale: green (AUPRC $\approx$ 0.5) to orange-red (AUPRC $\approx$ 1.0). (B) AUPRC for IBD, IBS, and Phylo in observed (solid dots) versus simulated (shaded interval) data. Horizontal dashed line marks the random classifier baseline. (C) IBD proportion density distributions for same-household (orange; n = 158) and different-household (grey; n = 12,562) pairs in the observed dataset. The vertical dashed lines indicate the median IBD proportion for each group.*

Discussion

This study set out to map when genomic data can and cannot detect malaria transmission, and the answer is unambiguous: in most biologically relevant conditions simulated here, reliable discrimination of transmission-linked parasite pairs from unlinked ones remains beyond the reach of current methods. AUPRC rarely exceeded 0.65 under any combination of method, recombination rate, migration, or sample size, and performance for *Plasmodium falciparum* relevant parameters particularly low recombination and high background IBD was substantially worse, approaching the random classifier baseline in the observed dataset. These findings are broadly consistent with the theoretical expectation that parasites with high rates of selfing and low effective recombination will maintain extensive identity by descent across unrelated transmission chains, blurring the genomic boundary between linked and unlinked pairs (Neafsey, Waterhouse, et al. 2015; Manske et al. 2012). However, the degree to which this boundary is irrecoverable even with ideal IBD-based inference was not previously quantified in a simulation framework that jointly varied recombination, population structure, and temporal transmission depth.

The dominance of IBD-based inference over IBS and phylogenetic approaches across essentially all conditions is consistent with prior work demonstrating that haplotype-level sharing provides more precise relatedness estimates than allele-by-allele identity for highly selfing organisms (Henden et al. 2018; Apinjoh et al. 2019). Schaffner et al. established hmIBD as the reference tool for pairwise IBD estimation in *P. falciparum* and noted that inference quality degrades with increasing recombination (Schaffner et al. 2018), a pattern directly replicated here in the declining correlation between true and inferred IBD across **Figure 3**. Chang et al. similarly showed that genetic relatedness measures could distinguish recent from older transmission in low-transmission settings, but their analyses were conducted in populations with relatively sparse background relatedness (Auburn et al. 2018). Our results suggest that this finding is not readily generalisable to high-transmission, high-IBD settings: when background relatedness is elevated, as in the observed dataset (**Figure 9C**), even true IBD provides insufficient discriminatory power, irrespective of inference accuracy. This represents a biological, not methodological, ceiling on detectability.

The finding that recombination rate was the dominant driver of transmission detectability exceeding

the effects of migration rate, sample size, and G threshold in most conditions extends and refines the theoretical framework established in population genetics of malaria. Nkhoma et al. demonstrated that recombination breaks down extended haplotype structure in *P. falciparum*, reducing IBD block length and eroding genealogical signal over time (Nkhoma et al. 2012). Our results place this erosion in a quantitative performance framework: the transition from $r = 10^{-9}$ to $r = 10^{-6}$ per base per generation translated to a gain of 0.22–0.30 AUPRC units across methods (**Figure 8**), far larger than any variation attributable to population structure or sample size. This has practical consequences. Genomic epidemiology programmes that benchmark tools using high-recombination reference datasets that implicitly assume recombination rates comparable to other pathogens may substantially overestimate detectability in the field. The WHO supported adoption of genomic surveillance for malaria (Didelot, Gardy, and Colijn 2014) needs to be grounded in explicit estimates of local recombination parameters before performance expectations can be set.

The role of the generation threshold G as a modulator of detectability has not previously been explored systematically in the malaria genomics literature, yet our results show it to be a first order determinant of performance. At $G = 1$, operationally the most relevant definition for outbreak or household transmission investigations, AUPRC was below 0.50 for all methods and all recombination rates. This is sobering. The phylogenetic transmission inference literature in bacterial pathogens, particularly tuberculosis and SARS-CoV-2, has broadly assumed that phylogenetic proximity is a reasonable proxy for recent transmission (Didelot, Gardy, and Colijn 2014; De Maio et al. 2015). In those organisms, mutation rates are orders of magnitude higher and recombination is absent or infrequent, making temporal signal recoverable from sequence data alone. In *P. falciparum*, neither condition holds. The TMRCA overlap between direct and non-direct pairs observed in **Figure 2** is a direct consequence of the organism’s biology, and the G -threshold analysis (**Figure 8**) confirms that the genomic signal necessary to resolve very recent transmission is simply not present at any recombination rate in the parameter space we explored, a finding with no clear analogue in the bacterial or viral transmission inference literature (Mohd Abd Razak et al. 2016).

Beyond benchmarking, these results carry implications for the theoretical framing of genomic transmission inference as an identifiability problem. In the formal sense, a statistical inference problem

is identifiable if distinct parameter values produce distinct observable distributions. The TMRCA overlap in **Figure 2** and the near-superimposable IBD distributions in **Figure 9C** together suggest that, under realistic malaria transmission conditions, the distributions of genomic distance between direct and non-direct pairs are not reliably distinguishable. This constitutes a form of practical non-identifiability that precedes any modelling assumption about transmission chains. Didelot et al. formalised similar concerns in the context of bacterial whole-genome sequence phylogenetics, arguing that without strong epidemiological priors, transmission reconstruction is severely underdetermined (Didelot, Siveroni, and Volz 2020). The malaria case is arguably more extreme because the genomic signal itself, not merely its interpretation, is contaminated by background relatedness and short IBD blocks. Moving forward, frameworks that integrate genomic data with spatial, temporal, and household-level epidemiological data are likely to be necessary to approach reliable transmission inference in high transmission settings (@ Wesolowski et al. 2018; Hay et al. 2002).

Limitations and scope

Several limitations of this study should be considered when interpreting these findings. First, simulations were restricted to a single chromosome (chromosome 1, 640,851 bp) of the *P. falciparum* genome rather than the full 14 chromosomes, with substantial heterogeneity in recombination rate, GC content, and functional constraint across the genome. In particular, telomeric and subtelomeric regions harboring hypervariable antigen gene families (var, rifin, stevor) violate the neutral evolution assumptions of both the simulation model and hmmIBD, and are typically excluded from relatedness analyses in practice. Chromosomes also differ in effective length and recombination landscape, which will produce chromosome-specific AUPRC values around the genome-wide average reported here. While chromosome 1 provides a tractable proof-of-concept framework, IBD statistics computed from the whole genome are expected to be more precise due to greater averaging over independent chromosomal segments. Whole-genome simulations would likely increase inferred IBD accuracy and may modestly raise the detectability ceiling, though the biological signal constraints identified here, particularly TMRCA overlap, would remain.

Second, the model does not incorporate within-host diversity (i.e., complexity of infection or multi-

plicity of infection, COI/MOI). Although COI was not explicitly modelled, the recombination sweep scenario partially captures its genomic consequences, as higher recombination rates mimic the increased haplotype fragmentation and reduced IBD block length that arise when multiple co-infecting clones contribute to the genomic background of a sampled parasite. Mixed infections are common in high-transmission settings and introduce substantial uncertainty into pairwise relatedness estimates, as hmmlBD and IBS-based tools perform less reliably on polyclonal samples [Henden et al. (2018); 15]. Our simulations therefore likely overestimate achievable field performance, as they implicitly assume monoclonal, correctly phased haplotypes. Third, the bottleneck parameterisation, while varied in sensitivity analysis (**Figure S2**), does not capture the full complexity of demographic history in endemic populations, including recent range expansion, seasonal bottlenecks, or the impact of malaria control interventions on N_e (Amambua-Ngwa et al. 2019). Fourth, the study lacks a formal identifiability analysis in the mathematical sense; the empirical AUPRC approach quantifies discriminative performance but does not derive analytical bounds on what is recoverable from the data. Finally, there is a potential circularity between the simulation model and inference: hmmlBD was designed with assumptions about recombination and selfing rates that partially overlap with those used to generate the simulation data, which may produce modestly optimistic IBD inference performance relative to a truly independent generative model.

Future Research Directions

Several directions follow directly from these findings. Whole-genome simulations incorporating all 14 *P. falciparum* chromosomes, including chromosome-specific recombination rates and the hypervariable subtelomeric regions, are a necessary next step to assess whether detectability limits identified here are specific to the single-chromosome setting or hold genome-wide. Incorporating polyclonal infection models with realistic COI distributions would substantially improve the biological fidelity of the simulation framework and is likely to reveal additional performance losses not captured here. Genome-wide simulation extending across all 14 chromosomes with empirical recombination maps (Miles et al. 2016; MalariaGEN) represents an important direction for future work, and would permit direct calibration of the recombination-driven identifiability ceiling reported here

against the full genomic architecture of *P. falciparum*. Methods that jointly model genomic relatedness, spatial distance, and household structure, as demonstrated in principle in some bacterial and viral settings (Wesolowski et al. 2018; Hay et al. 2002), should be adapted and evaluated for *P. falciparum*, with explicit attention to the high-relatedness, low-recombination regime that characterises endemic African populations. Empirical validation studies using datasets where household or clinic-confirmed transmission chains are known with higher certainty than household membership alone would also allow sharper calibration of the detectability limits mapped here. Lastly, formalising the non-identifiability conditions described in this paper, deriving analytical bounds on the minimum recombination rate, sample size, or temporal resolution required for transmission inference to achieve a specified precision, would provide the theoretical foundation currently missing from the malaria genomic epidemiology literature and could inform the design of future surveillance systems before costly sequencing investments are made.

Conclusions

This study set out to define the conditions under which genomic data can and cannot detect malaria transmission, using a simulation-based framework that jointly varied recombination rate, population structure, migration, sample size, and the temporal scope of the transmission definition. The central finding is that transmission detectability in *P. falciparum* is primarily constrained by recombination rate, with AUPRC rising substantially across the four order of magnitude range explored, and that no method, IBD, IBS, or phylogenetic inference, achieved reliable discriminative performance in the low-recombination, high-background relatedness regime most characteristic of endemic African settings. The generation threshold G emerged as a critical secondary determinant, with performance below chance for all methods at $G = 1$, the operationally most relevant temporal scope for outbreak and household transmission investigations. Validation against observed *P. falciparum* data confirmed that current tools are unlikely to resolve transmission links in high-transmission settings: the near-identical IBD distributions between household-linked and unlinked pairs indicate a biological signal ceiling that inference tools cannot overcome.

These findings reframe the problem of transmission inference in malaria as one of biological con-

481 straint rather than methodological inadequacy. The TMRCA overlap between direct and non-direct
482 transmission pairs observed in simulation is not an artefact of tool design but a consequence of the
483 organism's high selfing rate, high background relatedness, and short IBD blocks under endemic
484 transmission intensity. In this sense, the study contributes a formal, quantitative dimension to a
485 concern that has been raised qualitatively in the genomic epidemiology literature but not previ-
486 ously tested across the parameter space that governs field performance. The results argue for
487 a reorientation of expectations: genomic data in malaria may be most informative not as a stan-
488 dalone transmission discriminator, but as one layer within an integrated inference framework that
489 incorporates spatial proximity, temporal co-occurrence, and household or clinical contact data.

490 As genomic surveillance programmes expand across sub-Saharan Africa and Southeast Asia, the
491 question of what parasite genomes can reasonably tell us about transmission is no longer theoret-
492 ical. This study provides a quantitative answer for a defined parameter space and demonstrates
493 that the answer is sobering in most endemic conditions. Moving forward, the framework developed
494 here, simulation-based detectability mapping across biological parameter gradients, offers a tem-
495 plate for pre-deployment evaluation of genomic surveillance tools in any pathogen system where
496 background relatedness, recombination, and population structure jointly shape the feasibility of
497 transmission inference. Translating these insights into surveillance design, sampling strategies,
498 and analytical pipelines that are appropriately calibrated for the local biological context will be es-
499 sential to ensuring that genomic investments in malaria control deliver actionable epidemiological
500 intelligence.

Methods

Overview of the simulation framework

We developed a comprehensive simulation framework to evaluate the accuracy of genomic methods for reconstructing *Plasmodium falciparum* transmission networks. Our approach integrates forward-time population genetic simulation using SLiM v4.0 (Haller & Messer, 2023) with coalescent-based genealogical reconstruction using msprime v1.2 (Kelleher et al., 2016) and tree sequence recording via pyslim v1.0 (Haller et al., 2019). This hybrid framework allows us to capture both the selective dynamics of recent parasite evolution and the deeper coalescent history while maintaining computational tractability for genome-scale simulations.

We simulated three distinct scenarios to evaluate transmission inference under varying evolutionary conditions:

1. **Baseline scenario:** Fixed *P. falciparum* specific parameters representing typical endemic transmission
2. **Recombination sweep:** Systematic variation of recombination rates to assess the impact of within-host diversity on transmission reconstruction.
3. **Full parameter space:** Factorial exploration of recombination rate, transmission bottleneck size, mutation accumulation (expected substitutions per transmission, EST), sampling intensity, outbreak size, and population structure with migration.

Biological Model and Assumptions

Parasite Life Cycle and Reproductive Mode

P. falciparum exhibits a complex haploid-diploid life cycle alternating between human and mosquito hosts. Our simulations explicitly model this biology with the following key assumptions:

Haploidy: The parasite genome remains haploid throughout asexual replication in both human (liver and blood stages) and mosquito (sporozoite) hosts. Diploidy occurs only transiently during sexual reproduction in the mosquito midgut (zygote/ookinete stage), followed immediately by

meiosis. We therefore model parasites as fundamentally haploid organisms (one genome copy per individual) with occasional outcrossing.

High selfing rate: Following empirical estimates from endemic populations (Daniels et al., 2015), we set the selfing rate $\sigma = 0.95$ for baseline simulations. This reflects the predominantly clonal transmission of *P. falciparum* due to: low multiplicity of infection (MOI) in many endemic settings, temporal asynchrony of gametocyte production and stochastic sampling of gametocytes during mosquito blood meals.

The effective outcrossing rate is thus $\tau = 1 - \sigma = 0.05$.

Discrete generations: Each complete transmission cycle (human $\rightarrow$ mosquito $\rightarrow$ human) represents one discrete generation in our model. This comprises: - Human liver stage: 7-10 days (1 generation) - Human blood stage: 5 asexual replication cycles over ~ 10 days (5 generations) - Mosquito sexual stage: Gametocyte $\rightarrow$ ookinete (1-2 generations) - Mosquito asexual stage: Oocyst sporogony (8-10 generations) - Mosquito sporozoite migration: (1-2 generations)

The total generation time per transmission event is approximately $g_{\text{trans}} = 22$ generations (range: 18-28 depending on infection duration), spanning 12-15 days of calendar time.

Transmission Bottleneck

A critical feature of *P. falciparum* transmission is the severe genetic bottleneck occurring during mosquito-to-human transmission. When an infected mosquito takes a blood meal, only a small number of sporozoites successfully invade hepatocytes and establish blood-stage infection. We model this bottleneck explicitly with parameter B , representing the number of parasite genomes transmitted per infection event.

We explore three bottleneck scenarios: - **Tight bottleneck:** $B = 1$ (single parasite transmission, severe genetic drift) - **Moderate bottleneck:** $B = 5$ (few parasites, intermediate drift) - **Loose bottleneck:** $B = 20$ (multiple parasites, reduced drift)

Empirical estimates suggest $B \approx 1 - 10$ in natural transmission (Schneider et al., 2018), making this a key parameter governing the correlation between genetic and epidemiological relationships.

Population Structure and Demography

Single-population (baseline) model: We simulated a single panmictic population with an ancient effective population size: $N_e = 10,000$ (representing pre-bottleneck ancestral diversity) - Recent effective population size: $N_0 = 1,000$ (representing contemporary population) - Exponential population size change occurring at generation $g_{\text{change}} = 200$ before present:

$$N(t) = N_0 \left(\frac{N_e}{N_0} \right)^{t/g_{\text{change}}}$$

where t represents generations before present ($0 \leq t \leq g_{\text{change}}$).

Multi-population (migration) model: For the full parameter space exploration, we simulated $n = 3$ populations (demes) with symmetric migration at rate m per generation. The migration matrix $\mathbf{M}$ is:

$$M_{ij} = \begin{cases} 1 - (n - 1)m & \text{if } i = j \\ m & \text{if } i \neq j \end{cases}$$

We explored three migration scenarios:

- Low gene flow: $m = 0.001$ (1 migrant per 1000 generations)
- Moderate gene flow: $m = 0.01$ (1 migrant per 100 generations)
- High gene flow: $m = 0.05$ (1 migrant per 20 generations)

This represents varying degrees of spatial structure from nearly isolated populations to high connectivity, reflecting different endemic transmission settings (isolated villages vs. urban transmission hotspots).

Evolutionary Parameters

Recombination

Recombination in *P. falciparum* occurs during meiosis following sexual reproduction in the mosquito midgut. The recombination rate per base pair per generation, r , determines the extent to which parental genomes are reshuffled during outcrossing events.

We systematically varied r across four orders of magnitude:

- Very low: $r = 10^{-9} \text{ bp}^{-1} \text{ gen}^{-1}$ (effectively clonal)
- Low: $r = 10^{-8} \text{ bp}^{-1} \text{ gen}^{-1}$ (rare recombination)
- Medium: $r = 10^{-7} \text{ bp}^{-1} \text{ gen}^{-1}$ (moderate recombination)
- High: $r = 10^{-6} \text{ bp}^{-1} \text{ gen}^{-1}$ (frequent recombination)

Because direct simulation of polyclonal infections was beyond the scope of the current framework, effective recombination rate was used as a proxy for within-host complexity: elevated recombination rates approximate the increased outcrossing and haplotype mixing that occur when multiple genetically distinct clones co-infect a single host, thereby capturing, in aggregate, the signal erosion associated with higher multiplicity of infection.

For a chromosome of length $L = 640,851 \text{ bp}$ (chromosome 1), the genome-wide recombination rate is $R = 4N_e r L$, ranging from effectively no recombination ($R \approx 0.03$ for $r = 10^{-9}$) to free recombination ($R \approx 26$ for $r = 10^{-6}$).

Mutation

Two mutation processes were modeled as follow:

Neutral mutations: Added post-simulation using msprime with rate $\mu = 10^{-8} \text{ bp}^{-1} \text{ gen}^{-1}$, consistent with empirical estimates for *P. falciparum* (Hamilton et al., 2017). These mutations accumulate according to the infinite-sites model and provide genomic variation for IBD/IBS inference.

Selected mutations: To model adaptive evolution (drug resistance), we introduced a single beneficial allele at position $\ell_s = L/3$ with: - Selection coefficient: $s = 0.3$ (30% fitness advantage) -

593 Dominance coefficient: $h = 0.5$ (codominant, though relevant only during transient diploid stage)

594 - Introduction time: $g_s = 80$ generations before present

595 The selected allele was introduced into $n_{\text{origins}} = 1$ random genome and tracked until fixation
596 or loss. Simulations where the beneficial allele was lost were automatically restarted (maximum
597 100 restarts) to condition on successful selective sweeps, allowing us to evaluate transmission
598 inference in the presence of recent positive selection.

599 **Expected Substitutions per Transmission (EST)**

600 A key parameter linking mutation rate to transmission dynamics is the expected number of mu-
601 tations accumulated per transmission event. For a genome of length L transmitted over g_{trans}
602 generations with mutation rate μ :

$$\text{EST} = \mu \times L \times g_{\text{trans}}$$

603 We explored three EST scenarios:

- 604 • None: $\text{EST} = 0$ (mutations added only post-simulation via msprime)
- 605 • Low: $\text{EST} = 0.1$ (rare mutations during transmission)
- 606 • Moderate: $\text{EST} = 0.5$ (detectable mutation accumulation)

607 For $\text{EST} > 0$, we calculated the corresponding per-generation mutation rate as:

$$\mu_{\text{trans}} = \frac{\text{EST}}{L \times g_{\text{trans}}}$$

608 Using $L = 640,851$ bp and $g_{\text{trans}} = 22$ generations (typical scenario):

$$\mu_{\text{trans}} = \frac{\text{EST}}{640,851 \times 22} = \frac{\text{EST}}{14,098,722} \approx 7.1 \times 10^{-8} \times \text{EST}$$

609 For $\text{EST} = 0.5$: $\mu_{\text{trans}} \approx 3.5 \times 10^{-8} \text{ bp}^{-1} \text{ gen}^{-1}$.

Mathematical Framework

Wright-Fisher Model with Selection and Migration

Our simulations implement a discrete-generation Wright-Fisher model modified for haploid, selfing, and selection. The probability that genome i in generation $t + 1$ descends from genome j in generation t is:

$$P(i_{t+1} \leftarrow j_t) = \frac{w_j}{\bar{w}} \times \left[\sigma + (1 - \sigma) \times \frac{1}{N} \right]$$

where w_j is the fitness of genome j , $\bar{w} = \frac{1}{N} \sum_{k=1}^N w_k$ is the mean population fitness, σ is the selfing rate and N is the population size.

For the multi-population model, migration occurs before reproduction. The probability that genome i in population p at generation $t + 1$ descends from population q is M_{pq} , and given migration to population p , the probability of descending from a specific genome follows the within-population model above.

Fitness Model

For a genome carrying n copies of the beneficial allele at the selected locus (0 or 1 in haploids):

$$w = \begin{cases} 1 & \text{if } n = 0 \text{ (wild-type)} \\ 1 + s & \text{if } n = 1 \text{ (mutant)} \end{cases}$$

During transient diploidy (sexual stage), heterozygous genotypes have fitness $w = 1 + hs$, but this affects only the single generation of meiosis. The derived allele frequency in generation t , p_t , evolves deterministically (ignoring drift) as:

$$p_{t+1} = p_t \times \frac{1 + s}{1 + sp_t}$$

With drift in a finite population, frequency changes are stochastic, sampled from a binomial distribution:

$$X_{t+1} \sim \text{Binomial} \left(N, p_t \times \frac{1+s}{1+sp_t} \right)$$

where X_t is the count of derived alleles.

Transmission Bottleneck Model

Each transmission event samples B parasite genomes from the donor's infection. For a donor with derived allele frequency p_{donor} , the probability that the recipient infection has k copies of the derived allele is:

$$P(k \mid p_{\text{donor}}, B) = \binom{B}{k} p_{\text{donor}}^k (1 - p_{\text{donor}})^{B-k}$$

For tight bottlenecks ($B = 1$), transmission is deterministic: \

$$P(k = 1) = p_{\text{donor}}, P(k = 0) = 1 - p_{\text{donor}}$$

This bottleneck severely constrains the genetic diversity transmitted and creates strong stochasticity in allele frequency trajectories.

Coalescent Theory and Tree Sequences

The genealogical relationships among sampled parasites are recorded as tree sequences (Kelleher et al., 2018), encoding the full ancestral recombination graph (ARG). For neutral loci in a panmictic population of size N_e , the distribution of pairwise coalescence times T follows:

$$P(T > t) = e^{-t/(2N_e)}$$

with mean coalescence time $\mathbb{E}[T] = 2N_e$ generations.

Under the sequentially Markovian coalescent (SMC) approximation, genealogies at different genomic positions are correlated, with the correlation decaying as a function of recombination distance d :

$$\rho(d) = e^{-\rho d}$$

where $\rho = 4N_e r$ is the population-scaled recombination rate.

Tree sequences enable efficient computation of true **Identity by descent (IBD)** metric, which is the genomic segments sharing a common ancestor within T_{IBD} generations.

Evaluation

Because direct transmission events constitute a small fraction of all possible host pairs, we evaluate inference performance such as IBD, IBS and phylogenetic patristic distances using precision–recall curves and the area under the precision–recall curve (AUPRC), which is more informative than ROC-based metrics under class imbalance. Performance metrics were computed independently for each simulation replicate. For each recombination rate, results were summarized across replicates using the mean and interquartile range. Trends across recombination regimes were assessed using log-scaled recombination rates.

Simulation Workflow

Stage 1: Forward Simulation with SLiM

We used SLiM 4.0 to simulate $G = 200$ generations of forward-time evolution with the following workflow:

Initialization (generation 1):

- Create single population p_1 with $N = 10,000$ haploid individuals
- Initialize chromosome length $L = 640,851$ bp (chromosome 1)

- 663 • Set recombination rate r
- 664 • Enable tree sequence recording with pedigree tracking
- 665 • Define neutral mutation type m_1 and selected mutation type m_2

666 **Population size dynamics** (generations 1 to $200 - g_{\text{change}}$):

- 667 - Implement exponential growth/decline according to $N(t)$ equation
- 668 - For generation t :

$$N_t = N_0 \left(\frac{N_e}{N_0} \right)^{(200-t)/g_{\text{change}}}$$

669 **Selection initiation** (generation $200 - g_s - 1 = 119$):

- 670 - Save simulation state as checkpoint
- 671 - Introduce beneficial mutation at position ℓ_s in n_{origins} random genomes
- 672 - Track derived allele frequency (DAF) each generation

673 **Conditioning on establishment** (generations 119 to 200):

- 674 - Monitor derived allele frequency p_t each generation
- 675 - If $p_t = 0$ (loss) and restart count < 100 :
- 676 - Reload checkpoint state - Re-introduce mutation with new random seed
- 677 - Increment restart counter - If $p_t = 1$ (fixation): - End tracking (mutation has fixed) - Continue until
- 678 generation 200

679 **High relatedness simulation** (optional, final 50 generations):

- 680 - If `sim_relatedness = 1`, activate `modifyChild()` callback
- 681 - Filter mating pairs to prefer relatedness $1/32 < r_{ij} < 1/4$
- 682 - This enriches for specific relationship classes (half-siblings, cousins)
- 683 - Used to evaluate IBD methods on known pedigree relationships

684 **Output:**

- 685 - Tree sequence file (.trees) with complete genealogical history
- 686 - Derived allele frequency trajectory
- 687 - True effective population size (N_e) trajectory
- 688 - Restart count (number of failed establishment attempts)

689 **Stage 2: Sampling and Simplification**

690 From the final generation ($t = 200$), we sampled individuals to represent genomic surveillance:

691 **Sample selection:** - Identify all individuals alive at $t = 0$ (present day): $\mathcal{J}_{\text{alive}}$ - Randomly sample
692 $n_{\text{sample}}/2$ individuals from $\mathcal{J}_{\text{alive}}$ - Division by 2 because each haploid individual has 2 nodes in
693 tree sequence representation - Extract node IDs for sampled individuals - For multi-population
694 simulations, sample proportionally from each deme.

695 We varied sampling intensity from Low: 5% of outbreak ($n_{\text{sample}}/N_{\text{outbreak}} = 0.05$), Medium: 10-
696 20% of outbreak and High: 30% of outbreak (approaching exhaustive sampling).

697 **Tree sequence simplification:** We Reduce tree sequence to sampled lineages: `ts.simplify(sample_nodes)`,
698 remove non-ancestral branches (grafting lineages that don't coalesce to samples) and retain
699 input roots to preserve pre-simulation history. This reduces file size and computational burden for
700 downstream analyses

701 **Stage 3: Recapitation and Mutation Addition**

702 Forward simulation in SLiM captures recent evolutionary dynamics but does not extend to com-
703 plete coalescence (infinite time). We used msprime to “recapitate” the tree sequence, adding
704 ancient genealogical history. This extends all uncoalesced lineages backward in time under a
705 neutral Wright-Fisher model until complete coalescence, ensuring every genomic position has a
706 most recent common ancestor (MRCA). After recapitation, we removed all SLiM-generated muta-
707 tions (which were used only for selection tracking) and added neutral mutations uniformly across
708 the genome. Mutations are placed on tree branches according to a Poisson process with rate μ ,
709 generating realistic patterns of standing variation and derived polymorphisms. This allowed us to

710 generate a recapitated tree sequence, mutated tree sequence with final variant set and VCF file
711 with pseudo-homozygous genotypes used for the downstream analyses.

712 **Parameter Space Exploration**

713 We simulated three nested experimental designs:

714 **Scenario 1: Baseline (Fixed Parameters)**

715 Single replicate with *P. falciparum* typical parameters: - $N_e = 10,000$, $N_0 = 1,000$ - $r =$
716 $6.67 \times 10^{-7} \text{ bp}^{-1} \text{ gen}^{-1}$ (intermediate recombination) - $\mu = 10^{-8} \text{ bp}^{-1} \text{ gen}^{-1}$ - $B = 5$
717 (moderate bottleneck) - $\sigma = 0.95$ (high selfing) - $s = 0.3$, $g_s = 80$ - $n_{\text{sample}} = 100$ - Single
718 population (no migration)

719 **Purpose:** Establish baseline accuracy of IBD/IBS/phylogenetic patristic methods under realistic
720 conditions.

721 **Scenario 2: Recombination Sweep**

722 Systematic variation of recombination rate while holding other parameters constant: -
723 $r \in \{10^{-9}, 10^{-8}, 10^{-7}, 10^{-6}\} \text{ bp}^{-1} \text{ gen}^{-1}$ - All other parameters as in Scenario 1 - 10
724 replicates per recombination rate - Total: $4 \times 10 = 40$ simulations

725 **Purpose:** Isolate the effect of recombination on transmission inference accuracy.

726 **Scenario 3: Full Parameter Space**

727 Factorial design crossing:

- 728 • **Recombination rate:** $r \in \{10^{-9}, 10^{-8}, 10^{-7}, 10^{-6}\}$ (4 levels)
- 729 • **Bottleneck size:** $B \in \{1, 5, 20\}$ (3 levels)
- 730 • **Expected substitutions/transmission:** $\text{EST} \in \{0, 0.1, 0.5\}$ (3 levels)

- **Sampling proportion:** $f \in \{0.05, 0.10, 0.20\}$ (3 levels)
- **Outbreak size:** $N_{\text{outbreak}} = 2000$ (fixed)
- **Number of populations:** $n_{\text{pop}} = 3$ (fixed)
- **Migration rate:** $m \in \{0.001, 0.01, 0.05\}$ (3 levels)

Core biological scenarios: $4 \times 3 \times 3 = 36$ combinations With sampling/migration variation:
 $36 \times 3 \times 3 = 324$ combinations With 20 replicates: $324 \times 20 = 6,480$ total simulations.

Purpose: Comprehensively map the parameter space where genomic methods can accurately infer transmission, identifying critical thresholds and interaction effects.

Output and Data Availability

For each simulation, we generated:

Primary outputs: - Tree sequence files (`.trees`): Complete ARG with metadata - VCF files (`.vcf.gz`): Biallelic SNPs with phased haploid genotypes - Ground truth transmission network: Known donor-recipient pairs from pedigree - Parameter files (`.json`): Complete simulation metadata

Diagnostic outputs: - SLiM stdout logs (restart counts, population size trajectories) - Recapitation success metrics - Mutation count summaries

Empirical dataset for distributional consistency validation

To assess whether the simulated performance bounds are consistent with empirical *P. falciparum* genomic data, we applied the three inference methods to whole-genome sequence data from 160 *P. falciparum* isolates collected in 2014 from ten villages in The Gambia (MRC Unit The Gambia at LSHTM, 2014–2015; Table SX). Samples were collected from households enrolled in a longitudinal malaria surveillance programme. Household membership (HHCode, VillageCode) and visit date were used to define epidemiological proximity. Because confirmed transmission ground truth is unavailable in this observational dataset, same-household pairs sampled within the same transmission season were used as a proxy for transmission-linked pairs, following the approach

756 of Daniels et al. (2015) and Taylor et al. (2017). IBD inference was restricted to monoclonal infec-
757 tions ($\text{COIL} = 1$; $n = 152$) to satisfy hmmlBD input assumptions. All three methods were applied
758 using the same pipeline as the simulation analyses, and AUPRC was computed using household
759 proximity as the outcome label.

760 All simulation codes, parameter files, and analysis scripts are available at [GitHub repository URL].

761 Simulated datasets are archived at [data repository DOI].

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910 **Supplementary Information**

911 **Acknowledgements**

912 **Author contributions**

913 **Funding**

914 **Data availability**

915 **Ethics approval and consent to participate**

916 **Competing interests**

917 The authors declare no competing interests

Supplementary Figures

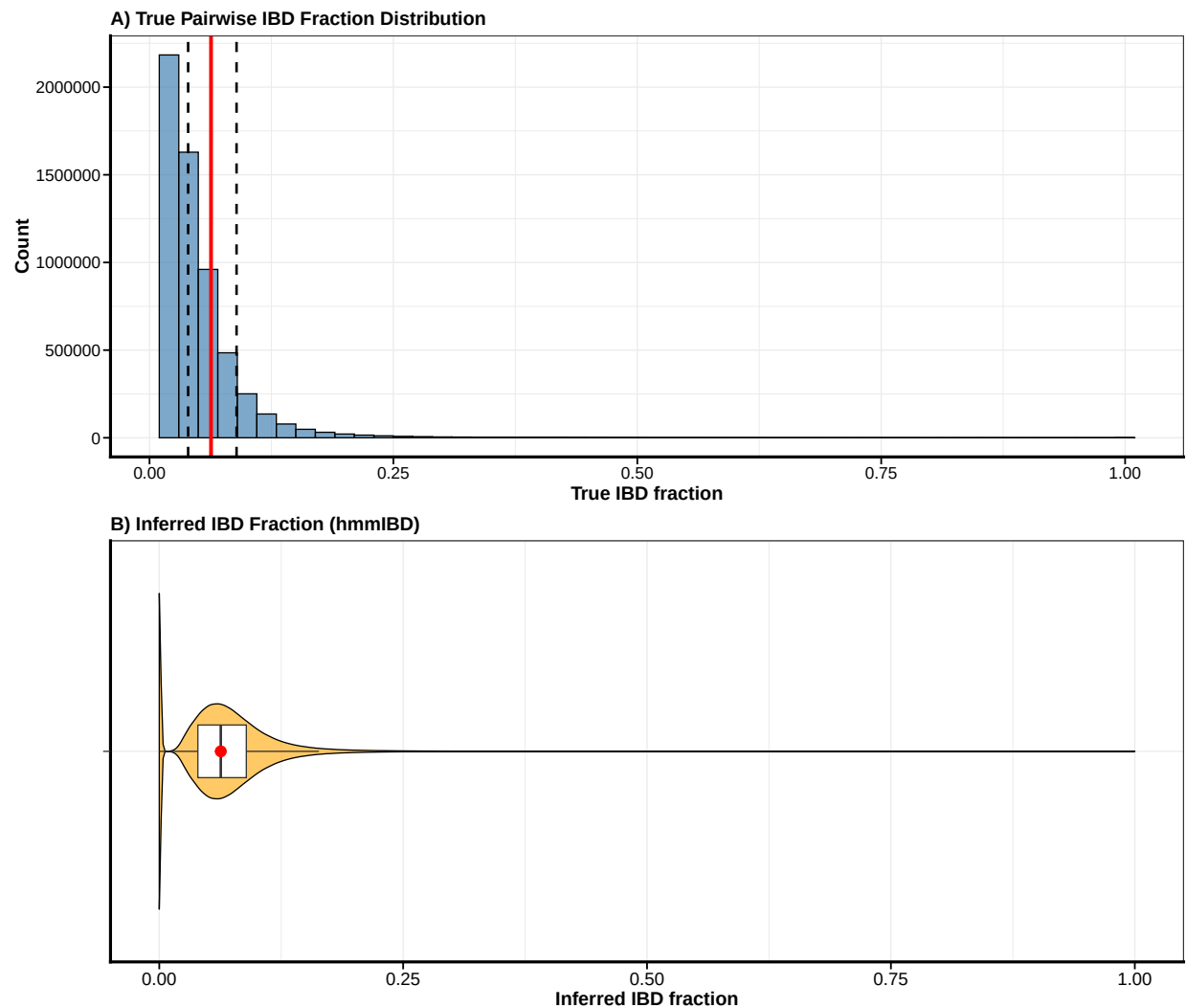


Figure S1. True pairwise IBD fraction distribution (panel A) and hmmIBD-inferred IBD fraction distribution (panel B) under the baseline scenario. Both panels show the full count distribution across all simulated pairs. The bimodal structure in panel A reflects the high-selfing, low-recombination baseline parameterization.

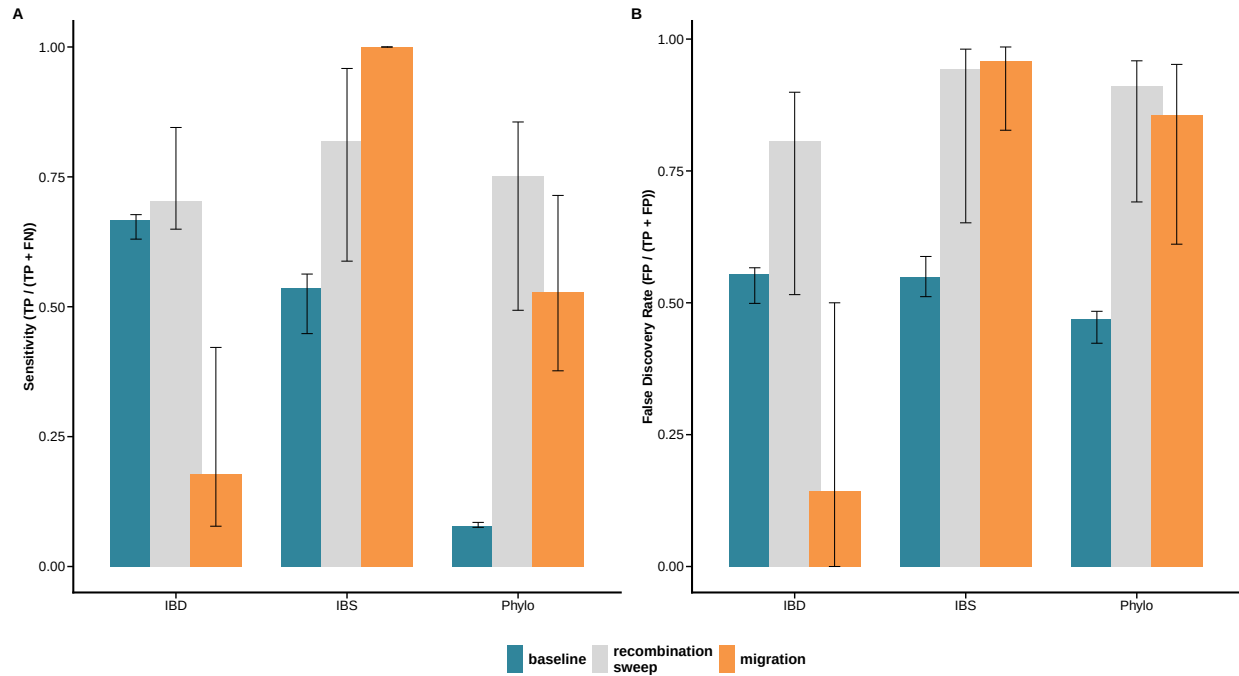


Figure Sx. Sensitivity (true positive rate; panel A) and false discovery rate (FDR = FP / (TP + FP); panel B) for IBD, IBS, and Phylo methods across the baseline, recombination sweep, and migration scenarios. Each dot and whisker represents the median and interquartile range across replicates. Results shown at a fixed classification threshold.

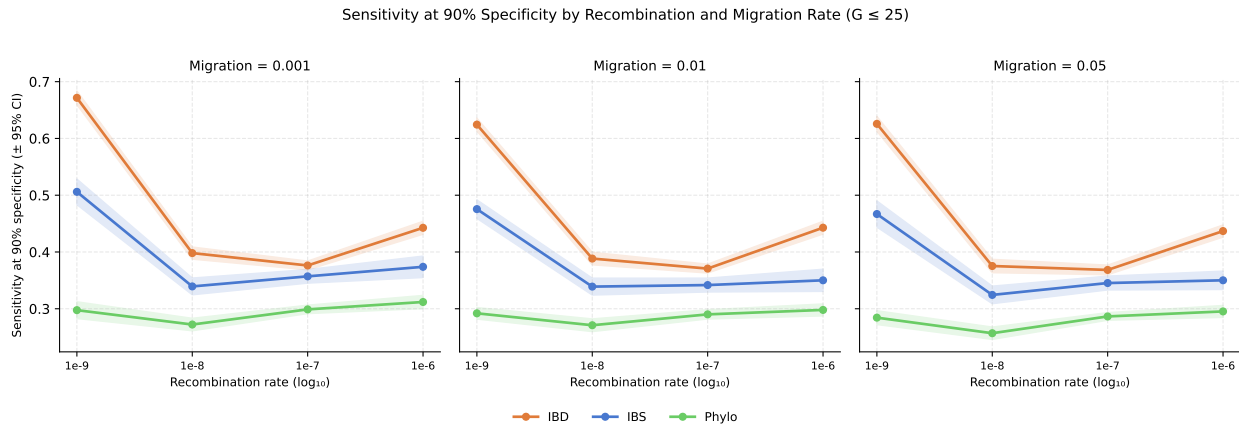


Figure S3. Sensitivity at 90% specificity (± 95% CI) for IBD (orange), IBS (blue), and Phylo (green) across four recombination rates (panels) and three migration rates (x-axis) for G ≤ 25 generations.

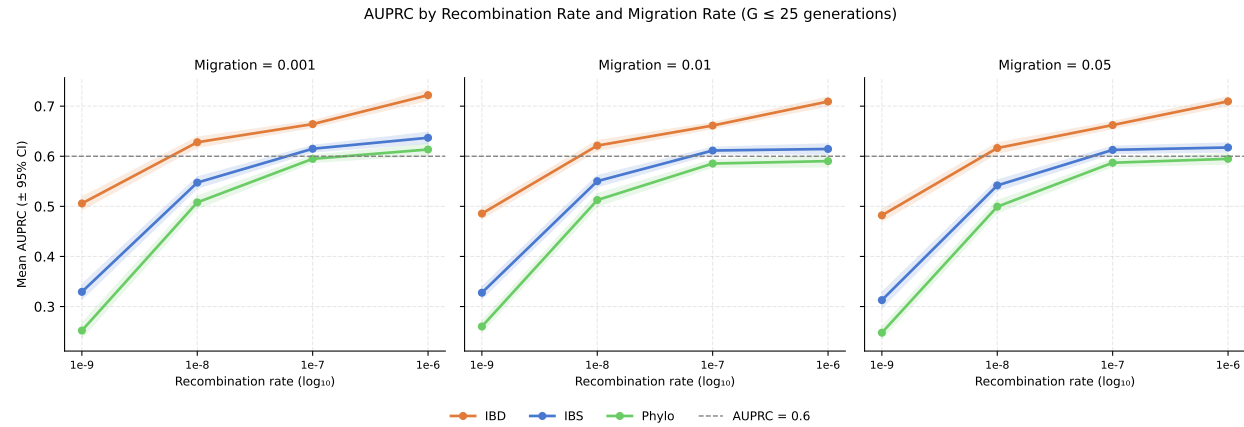


Figure S6. Mean AUPRC ($\pm$ 95% CI) for IBD, IBS, and Phylo across recombination rates (x-axis, log scale) for three migration rates (panels: 0.001, 0.01, 0.05), at $G \leq 25$ generations. A horizontal dashed reference line marks AUPRC = 0.6.

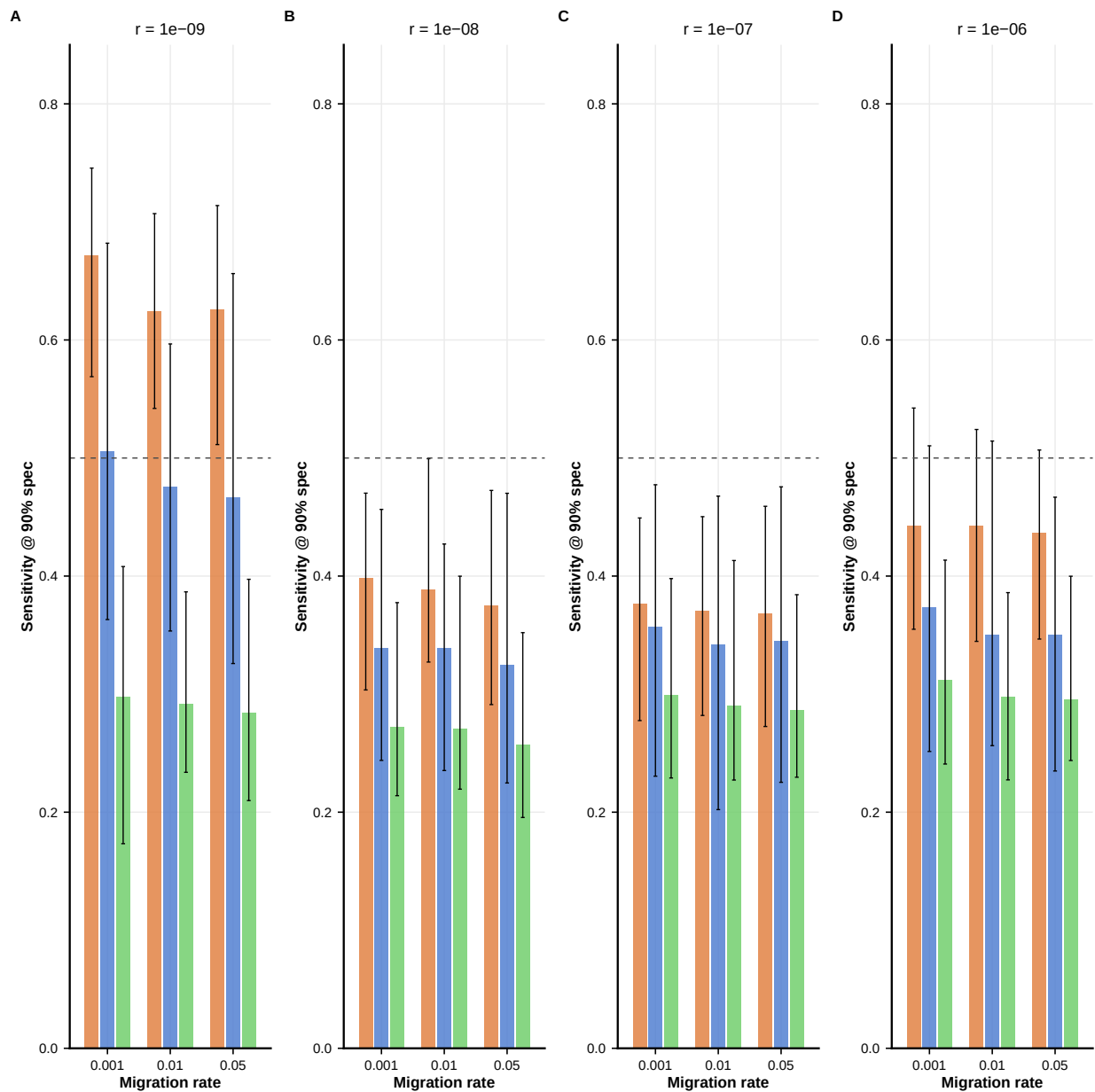


Figure Sxxx (migration sensitivity). Sensitivity at 90% specificity ($\pm 95\%$ CI) as a function of migration rate (x-axis: 0.001, 0.01, 0.05) for each recombination rate stratum (panels A–D: $r = 10^{-9}$, 10^{-8} , 10^{-7} , 10^{-6}). Colours indicate method (IBD, IBS, Phylo).

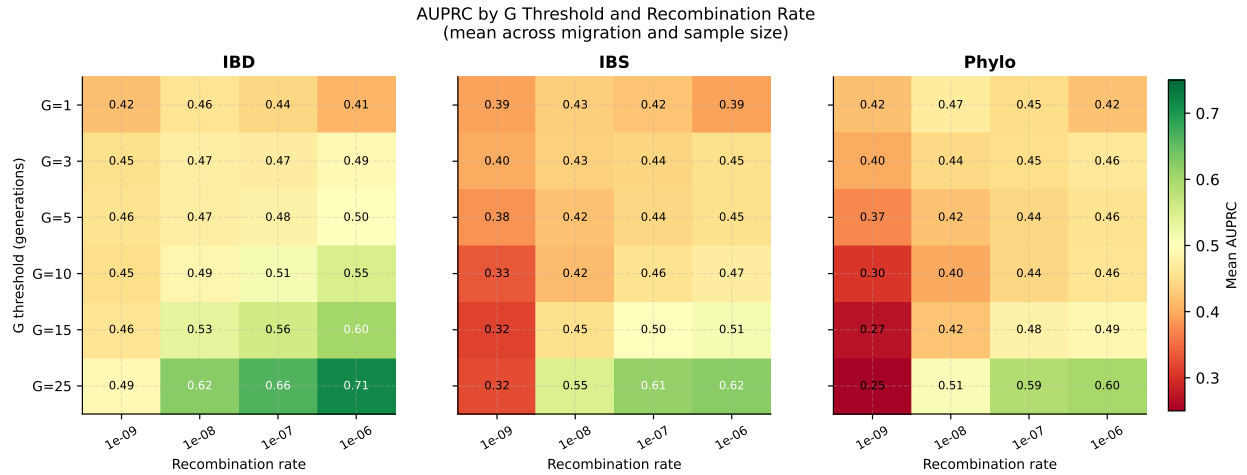


Figure S7. Full G-threshold × recombination rate AUPRC heatmaps for IBD, IBS, and Phylo methods (panels), mirroring the structure of Fig. 8A–C. Shown separately as a supplementary reference with extended numerical annotations.

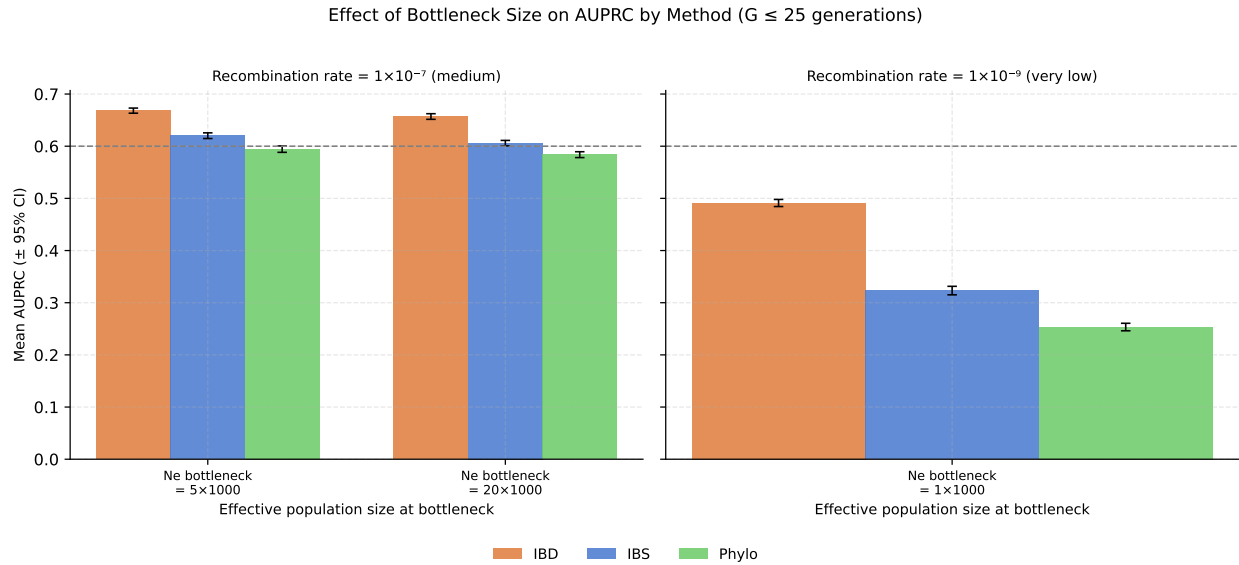


Figure S2. Effect of effective population size at the bottleneck (N_e bottleneck = 1,000; 5,000; 20,000) on mean AUPRC ($\pm 95\%$ CI) for IBD, IBS, and Phylo, shown separately for low ($r = 10^{-9}$) and medium ($r = 10^{-8}$) recombination rates. Results are for $G \leq 25$ generations.

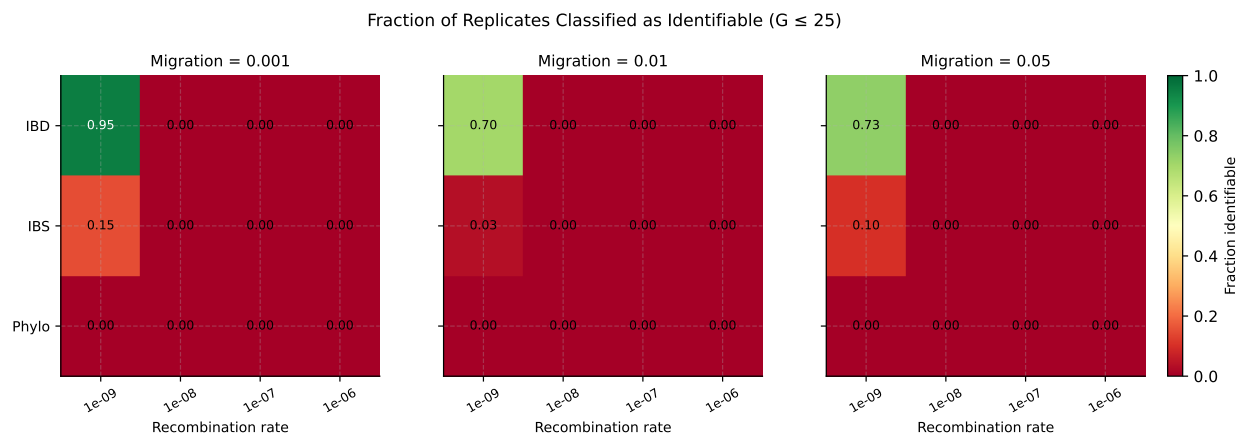


Figure S4. Fraction of simulation replicates classified as identifiable (i.e., achieving high-performance AUPRC above a fixed threshold) for IBD, IBS, and Phylo methods, across three migration rates (rows) and four recombination rates (columns). For $G \leq 25$ generations. Cells show the fraction (0–1); colour scale from dark red (fraction = 0) to dark green (fraction = 1).

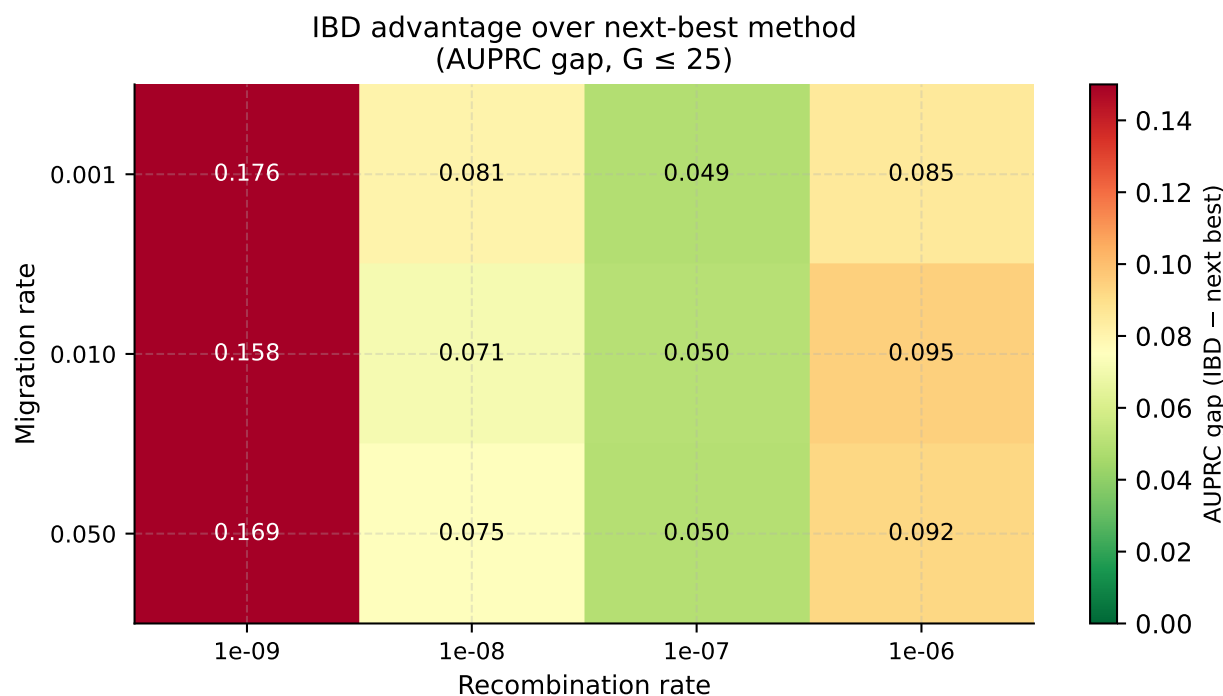


Figure S5. IBD advantage over the next-best method (AUPRC gap = $\text{AUPRC}_{\text{IBD}} - \max(\text{AUPRC}_{\text{IBS}}, \text{AUPRC}_{\text{Phylo}})$) for $G \leq 25$ generations. Heatmap with migration rate on the y-axis (0.001, 0.01, 0.05) and recombination rate on the x-axis (10^{-9} , 10^{-8} , 10^{-7} , 10^{-6}). Numeric values shown in each cell; colour scale from green (small gap) to dark red (large gap).

Potential Tables for the Manuscript

Table 1. Simulation parameters and scenario definitions A structured summary of all three scenarios (Baseline, Recombination Sweep, Multi-population), listing the values of μ , r , N_e , selfing rate, drug selection coefficient, number of subpopulations, migration rates, and the factorial combinations explored. This table contextualises all downstream results and is essential for reproducibility.

Table 2. AUPRC summary by method, scenario, and G threshold (main conditions) Mean AUPRC ($\pm$ 95% CI) for IBD, IBS, and Phylo at $G = 1, 5, 10$, and 25 , for each scenario and key recombination rate (e.g., $r = 10^{-9}$ and $r = 10^{-6}$). This table complements Fig. 8 and provides exact values for readers wishing to compare performance at specific conditions.

Table 3. Sensitivity at 90% specificity across recombination and migration (key conditions) Mean sensitivity at 90% specificity ($\pm$ 95% CI) for IBD, IBS, and Phylo at each recombination $\times$ migration combination for $G \leq 25$ generations. This provides a fixed-specificity operating point complement to the AUPRC-based summaries, facilitating comparison with threshold-based field application scenarios. Complements Fig. S3 and FigSxxx_migration_sensitivity.

Table 4. Observed versus simulated AUPRC and matched simulation conditions For each method (IBD, IBS, Phylo), the observed AUPRC in the real *P. falciparum* dataset, the range of simulated AUPRC values across all conditions, and the simulated condition (recombination rate, migration, sample size) that most closely matches the observed performance. Complements Fig. 9B and anchors the biological interpretation of the simulation results.